

A Supplementary figures

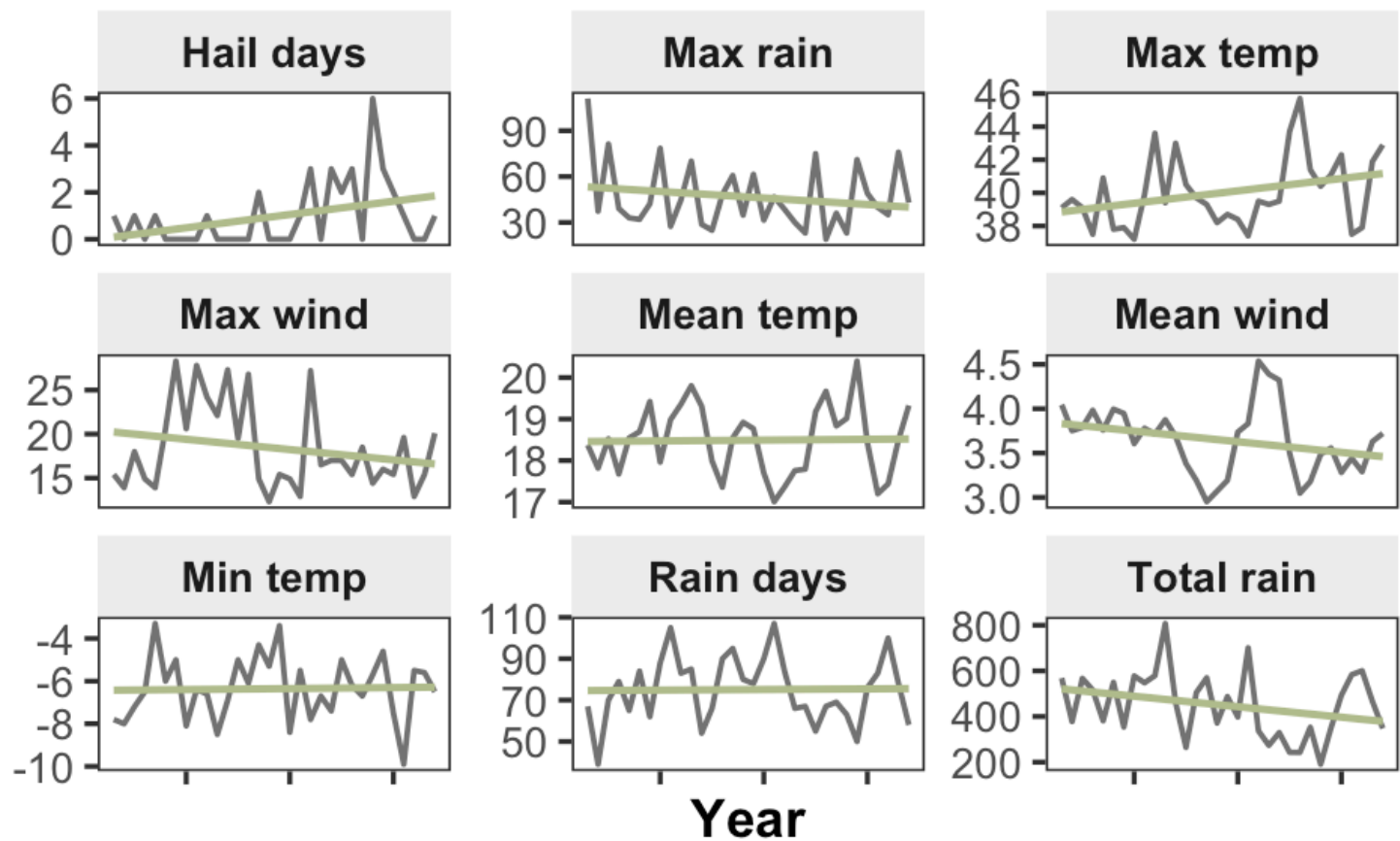


Figure A1: The temporal trends of the weather variables included in the analysis. Data is derived from the Kimberley meteorological station (1993 - 2024). Annual values are represented by the grey lines with the green lines representing the fitted linear trend.

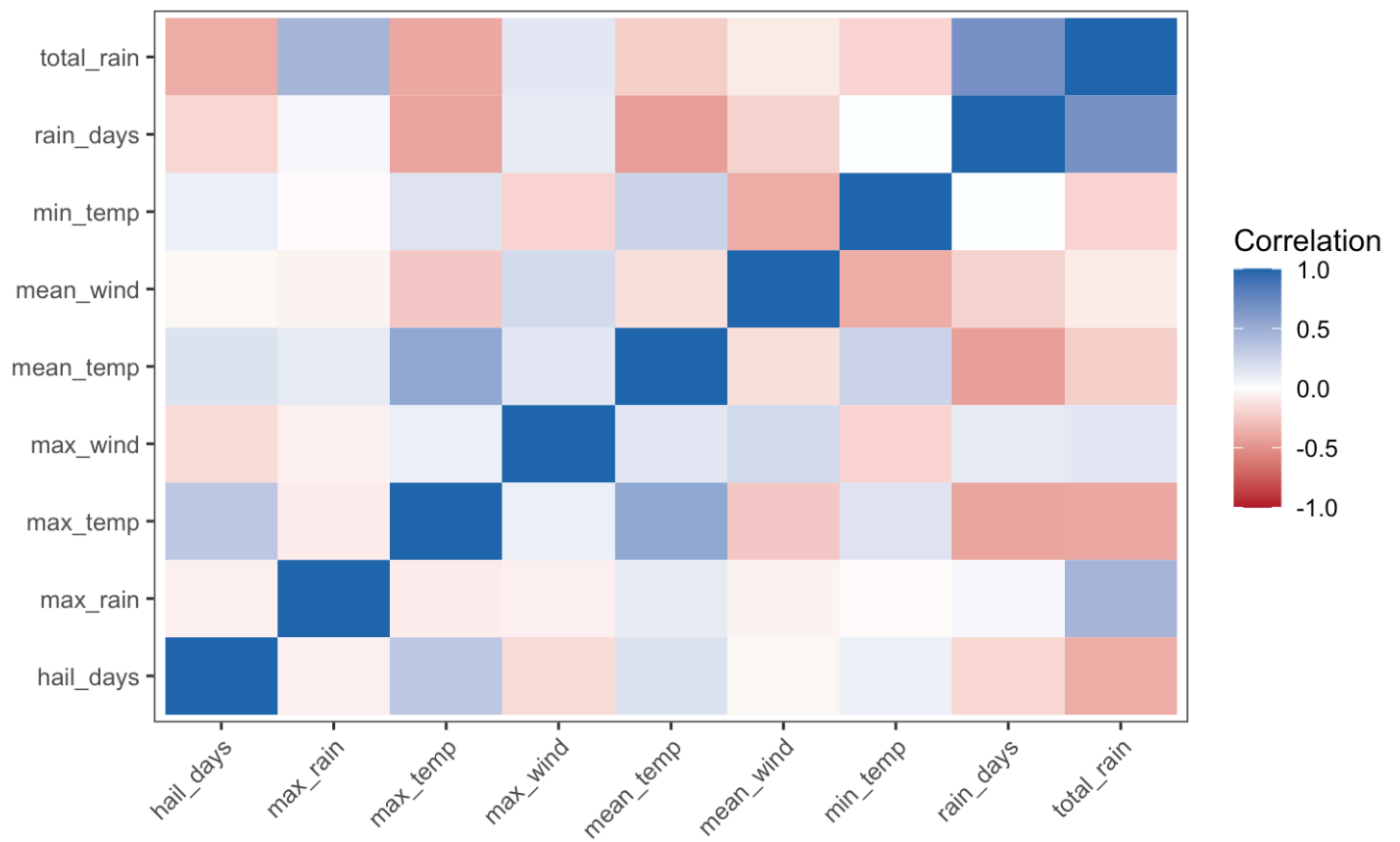


Figure A2: A correlation matrix showing the persons correlation coefficients between all weather variables included in the analyses on the effects of weather on the breeding ecology of African white-backed vultures at Dronfield (1993-2025).

Table 1: Codes used to describe nest and breeding status during field surveys at Dronfield Nature Reserve, Kimberley.

Code	Description
1	Chick ringed.
2	Chick too small to ring.
3	Chick too big to ring.
4	Chick fledged.
5	Nest new lining (for this breeding season).
6	Nest egg shell fragments (for this breeding season).
8	Nest with white wash.
10	Nest inactive.
11	Nest collapsed.
13	Nest gone.
14	Other bird species nesting.
15	Egg (for this breeding season).
16	Nest rebuilt, lined with dead chick, bones or feathers (for this breeding season).
17	Nest rebuilt, not lined (for this breeding season).
18	Nest under construction (for this breeding season).
19	Found in August, not checked in October.
20	Found in October, not seen in August.
–	Nest tree, used for the first time this year.
–	Pylon nest.

B Full statistical analyses

```
library(showtext)
library(sysfonts)
library(systemfonts)

# Find the actual font file on your system
mf <- systemfonts::match_font("Times New Roman")

# Register it for plotting
sysfonts::font_add(family = "Times", regular = mf$path)

# Turn on showtext so PDF devices embed it safely
showtext::showtext_auto(enable = TRUE)
```

B.1 Load data

```
vultures <- read.csv("data/93_24_cleaned.csv")
```

B.1.1 Clean breeding data

```
vultures.clean <- vultures |>
  clean_names() |>
  rename_with(~ gsub("_", ".", .x)) |>
  mutate(
    ringing.date = ringing.date |> as.character() |> str_trim(),
    ringing.date = if_else(ringing.date %in% c("", "NA", "?"), NA_character_, ringing.date),
    ringing.date = ymd(gsub("/", "-", ringing.date)),

    laying.date = laying.date |> as.character() |> str_trim(),
    laying.date = if_else(laying.date %in% c("", "NA", "?"), NA_character_, laying.date),
    laying.date = ymd(gsub("/", "-", laying.date))
  )
```

B.1.2 Explore data

```
glimpse(vultures.clean)
```

```
Rows: 2,657
Columns: 14
$ ringing.date      <date> 1993-09-21, 1993-09-21, 1993-09-19, 1993-09-1~
$ tree.drn          <chr> "1", "2", "3", "4", "5", "6", "8", "12", "13", ~
$ southings         <chr> " 28 38 43.8", "28 38 48.2", "28 38 55.6", "28~
$ eastings          <chr> "24 47 14.5", "24 47 05.1", "24 47 15.4", "24 ~
$ code              <int> 1, 1, 1, 1, 1, 1, 1, 15, 1, 1, 1, 1, 3, 1, 5, ~
$ ring              <chr> "G19791", "G19792", "G19781", "G19780", "G1978~
$ colour.rings.r.top.down <chr> "M", "M", "M", "M", "M", "M", "M", "", "M", "M~
$ colour.rings.l.top.down <chr> "B, Y, Y", "B, Y, R", "B, W, G", "B, B, R", "B~
$ mass.g            <int> 5900, 5000, 3900, 4950, 4950, 3800, 3550, NA, ~
$ wing.length.mm    <int> 475, 425, 260, 310, 290, 275, 210, NA, 245, 29~
```

```
$ age.at.ringing.days      <int> 89, 80, 57, 64, 61, 59, 49, NA, 55, 61, 5, 92,~
$ value.at.ringing        <int> 34233, 34233, 34231, 34231, 34231, 34231, 3423~
$ hatching.date          <chr> "1993/06/24", "1993/07/03", "1993/07/24", "199~
$ laying.date             <date> 1993-04-29, 1993-05-08, 1993-05-29, 1993-05-2~
```

```
summary(vultures.clean)
```

```
ringing.date      tree.drn      southings      eastings
Min.   :1993-09-19 Length:2657 Length:2657 Length:2657
1st Qu.:2004-06-06 Class :character Class :character Class :character
Median :2012-10-13 Mode  :character Mode  :character Mode  :character
Mean   :2011-08-31
3rd Qu.:2019-08-21
Max.   :2024-10-18
NA's   :32

code      ring      colour.rings.r.top.down
Min.   : 1.000 Length:2657 Length:2657
1st Qu.: 1.000 Class :character Class :character
Median : 1.000 Mode  :character Mode  :character
Mean   : 5.001
3rd Qu.: 6.000
Max.   :138.000
NA's   :4

colour.rings.l.top.down mass.g wing.length.mm age.at.ringing.days
Length:2657 Min.   : 2 Min.   : 25.0 Min.   : 5.00
Class :character 1st Qu.:4200 1st Qu.:320.0 1st Qu.: 65.00
Mode  :character Median :4800 Median :395.0 Median : 76.00
Mean   :4610 Mean   :375.7 Mean   : 74.18
3rd Qu.:5200 3rd Qu.:460.0 3rd Qu.: 86.00
Max.   :6700 Max.   :570.0 Max.   :300.00
NA's   :1225 NA's   :1192 NA's   :1198

value.at.ringing hatching.date laying.date
Min.   :34231 Length:2657 Min.   :1900-01-02
1st Qu.:37541 Class :character 1st Qu.:2002-06-02
Median :40824 Mode  :character Median :2011-05-13
Mean   :40489 Mean   :2010-05-21
3rd Qu.:43386 3rd Qu.:2018-05-26
Max.   :45583 Max.   :2024-08-11
NA's   :1231 NA's   :1197
```

```
colSums(is.na(vultures.clean))
```

```
ringing.date      tree.drn      southings
32                0            0
eastings          code          ring
0                4            0
colour.rings.r.top.down colour.rings.l.top.down mass.g
0                0            1225
wing.length.mm   age.at.ringing.days value.at.ringing
1192             1198            1231
hatching.date   laying.date
0               1197
```

B.1.3 Number of active nests

```
active_nests_yearly <- vultures.clean |>
  mutate(year = year(ringing.date)) |>
  filter(!is.na(year)) |>
  count(year, name = "active_nests")

ggplot(active_nests_yearly, aes(x = year, y = active_nests)) +
  geom_col(fill = "grey", alpha = 0.85) +
  labs(x = "Year", y = "Number of active nests") +
  theme_bw() +
  theme(panel.grid = element_blank())
```

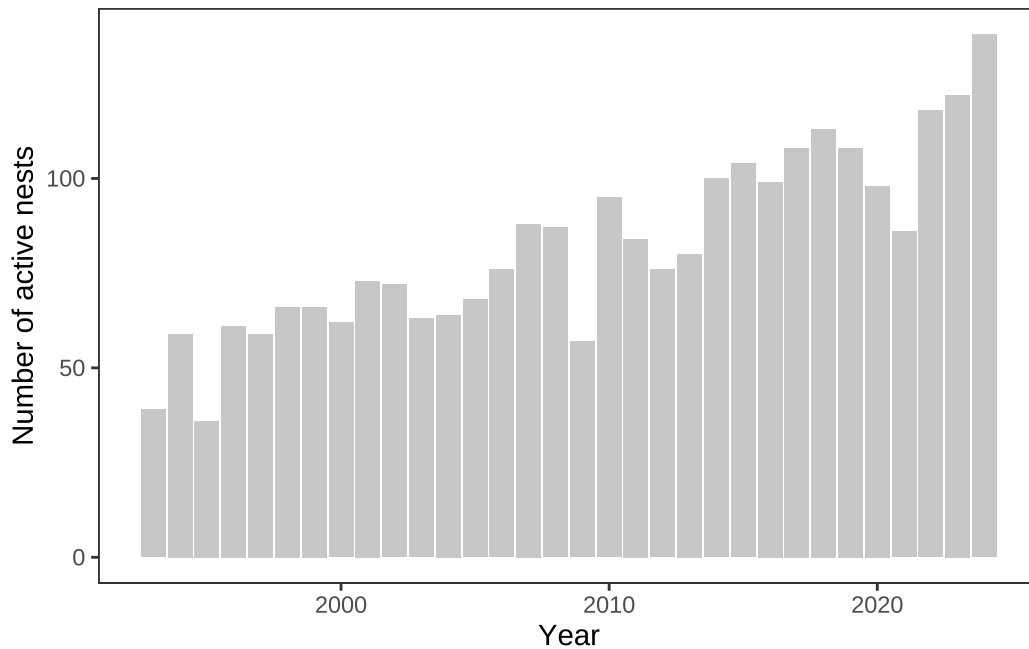


Figure 1: The number of active nests at Dronfield (1993–2024).

B.1.4 Breeding success

```
success_yearly <- vultures.clean |>
  mutate(year = year(ringing.date)) |>
  filter(!is.na(year)) |>
  summarise(
    active_nests = n(),
    success_nests = sum(!is.na(laying.date)),
    .by = year
  ) |>
  mutate(
    failed_nests = active_nests - success_nests,
    success_prop = success_nests / active_nests,
    fail_prop = 1 - success_prop
  ) |>
  filter(active_nests > 0)

ggplot(success_yearly, aes(x = success_prop)) +
  geom_histogram(bins = 15, fill = "grey") +
  labs(x = "Breeding success proportion", y = "Count") +
  theme_bw() +
  theme(panel.grid = element_blank())
```

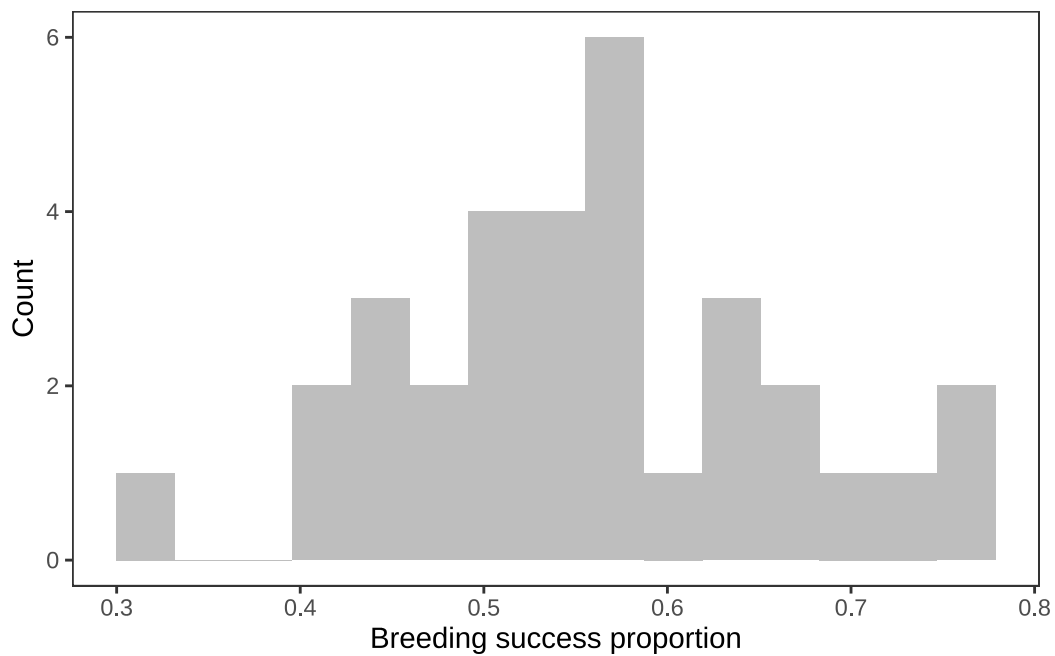


Figure 2: Histogram showing the distribution of annual breeding success proportions at Dronfield (1993–2024).

B.1.5 Failed vs Successful nests

```
mod_fail <- lm(fail_prop ~ active_nests, data = success_yearly)

newdat <- tibble(
  active_nests = seq(
    min(success_yearly$active_nests, na.rm = TRUE),
    max(success_yearly$active_nests, na.rm = TRUE),
    length.out = 200
  )
)

pr <- predict(mod_fail, newdata = newdat, interval = "confidence") |> as.data.frame()
pred_fail <- bind_cols(newdat, pr) |>
  rename(fit = fit, lwr = lwr, upr = upr)

ggplot() +
  geom_point(
    data = success_yearly,
    aes(x = active_nests, y = fail_prop),
    colour = "black",
    alpha = 0.7,
    size = 2.6
  ) +
  geom_ribbon(
    data = pred_fail,
    aes(x = active_nests, ymin = lwr, ymax = upr),
    fill = "#D5D6D2",
    alpha = 0.4
  ) +
  geom_line(
    data = pred_fail,
    aes(x = active_nests, y = fit),
    colour = "#D5D6D2",
    linewidth = 1.2
  ) +
  labs(x = "Number of active nests", y = "Proportion of nests failing") +
  theme_bw(base_family = "Times New Roman") +
```

```
theme(
  panel.grid = element_blank(),
  axis.title = element_text(face = "bold", size = 12),
  axis.text = element_text(face = "bold", size = 12)
)
```

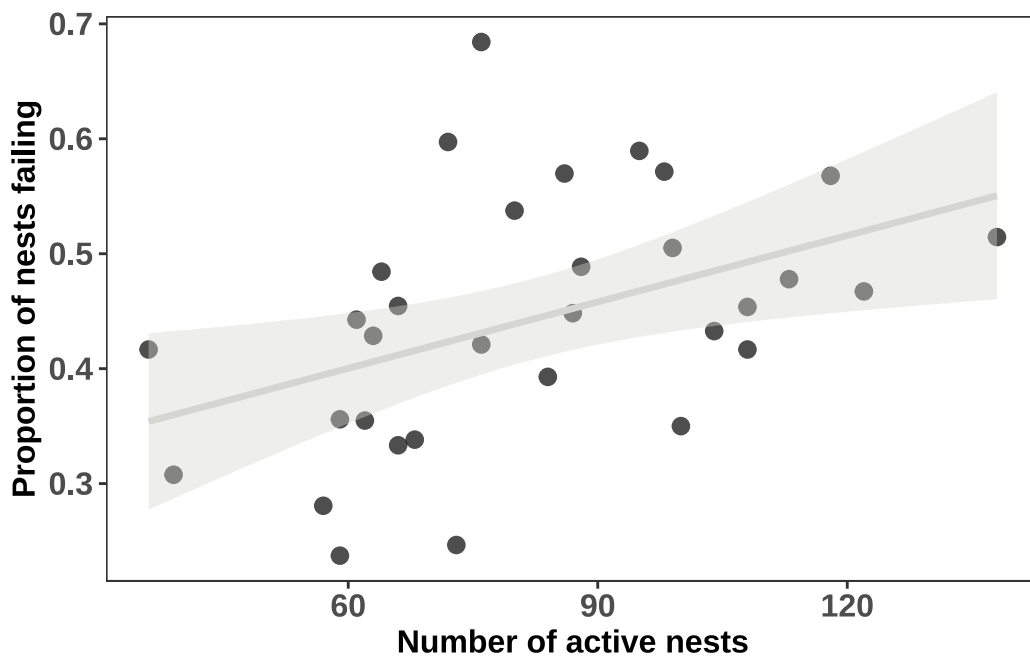


Figure 3: The proportion of failed nests vs successfull nests at Dronfield (1993-2024)

B.1.6 Weather data (GSOD)

```
options(timeout = 600)
options(download.file.method = "libcurl")

kim_station <- "684380-99999"

weather_raw <- NULL
for (i in 1:5) {
  message("GSOD download attempt: ", i)
  weather_raw <- try(
    GSODR::get_GSOD(station = kim_station, years = 1992:2024),
    silent = TRUE
  )
  if (!inherits(weather_raw, "try-error")) break
  Sys.sleep(2 * i)
}

if (inherits(weather_raw, "try-error")) {
  stop("GSOD download failed after 5 attempts. Run this chunk interactively or use cached data (see below).")
}

WeatherData <- weather_raw |>
  transmute(
    YEAR, MONTH, DAY,
    TEMP = as.numeric(TEMP),
    MAX = as.numeric(MAX),
    MIN = as.numeric(MIN),
    PRCP = as.numeric(PRCP),
    WDSP = as.numeric(WDSP),
```



```

MXSPD = as.numeric(MXSPD),
I_HAIL = as.numeric(I_HAIL)
) |>
arrange(YEAR, MONTH, DAY) |>
mutate(date = make_date(YEAR, MONTH, DAY))

```

B.1.7 Adding correct rainfall for 2004

```

lonlat <- tibble(lon = 24.81, lat = -28.62)

kimberley_chirps_2004 <- get_chirps(
  object    = lonlat,
  dates     = c("2004-01-01", "2004-12-31"),
  server    = "ClimateSERV",
  as.matrix = FALSE
)

daily_chirps_2004 <- tibble(
  date      = as.Date(kimberley_chirps_2004$date),
  rain_mm   = as.numeric(kimberley_chirps_2004$chirps)
)

annual_chirps_2004 <- daily_chirps_2004 |>
  mutate(YEAR = year(date)) |>
  summarise(
    total_rain_2004 = sum(rain_mm, na.rm = TRUE),
    max_rain_2004   = max(rain_mm, na.rm = TRUE),
    rain_days_2004  = sum(rain_mm > 0, na.rm = TRUE)
  )

annual_chirps_2004

```

```

# A tibble: 1 x 3
  total_rain_2004 max_rain_2004 rain_days_2004
      <dbl>         <dbl>         <int>
1         461.          28.8           54

```

```

WeatherData_clean <- WeatherData |>
  left_join(daily_chirps_2004, by = "date") |>
  mutate(
    PRCP = if_else(YEAR == 2004 & !is.na(rain_mm), rain_mm, PRCP)
  ) |>
  select(-rain_mm)

```

```

library(dplyr)
library(lubridate)
library(tibble)

# Daily weather
weather_daily <- WeatherData_clean |>
  mutate(
    date = as.Date(date),
    year = year(date)
  )

# Fixed window
weather_march_by_year <- weather_daily |>
  filter(month(date) == 3) |>      # March
  group_by(year) |>
  summarise(

```

```

mean_temp_win = mean(TEMP, na.rm = TRUE),
max_temp_win  = max(MAX,   na.rm = TRUE),
min_temp_win  = min(MIN,   na.rm = TRUE),
total_rain_win = sum(PRCP,  na.rm = TRUE),
max_rain_win   = max(PRCP,  na.rm = TRUE),
rain_days_win  = sum(PRCP > 0, na.rm = TRUE),
mean_wind_win  = mean(WDSP, na.rm = TRUE),
max_wind_win   = max(MXSPD, na.rm = TRUE),
hail_days_win  = sum(I_HAIL, na.rm = TRUE),
n_days         = n(),
.groups = "drop"
) |>
arrange(year)

weather_march_by_year

```

```

# A tibble: 33 x 11
  year mean_temp_win max_temp_win min_temp_win total_rain_win max_rain_win
<dbl>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1  1992         23.8         37         5.3        55.1        23.1
2  1993         21.0         31.6         5.8        12.7         7.62
3  1994         21.7         34.5         9.7        38.4        11.4
4  1995         22.1         35          9.4       155.        42.7
5  1996         22.3         34.7         8.2         3.55         2.03
6  1997         19.8         33.1         8.8       118.        33.0
7  1998         22.2         36.7          9       100.         32
8  1999         25.4         37.8        12.8        39.9        20.1
9  2000         22.3         33.2        12.4        89.6        28.2
10 2001         23.2         36        11.3        57.7        24.1
# i 23 more rows
# i 5 more variables: rain_days_win <int>, mean_wind_win <dbl>,
#   max_wind_win <dbl>, hail_days_win <dbl>, n_days <int>

```

B.1.8 Lagged weather summaries

```

weather_annual_by_year <- weather_daily |>
  group_by(year) |>
  summarise(
    mean_temp_annual = mean(TEMP, na.rm = TRUE),
    min_temp_annual  = mean(MIN, na.rm = TRUE),
    total_rain_annual = sum(PRCP, na.rm = TRUE),
    rain_days_annual  = sum(PRCP > 0, na.rm = TRUE),
    mean_wind_annual  = mean(WDSP, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(year)

weather_annual_by_year <- weather_annual_by_year |>
  mutate(
    lag_mean_temp_annual = lag(mean_temp_annual, 1),
    lag_total_rain_annual = lag(total_rain_annual, 1),
    lag_rain_days_annual  = lag(rain_days_annual, 1)
  )

summary(weather_annual_by_year)

```

```

year      mean_temp_annual min_temp_annual total_rain_annual

```

Min.	:1992	Min.	:17.01	Min.	: 8.195	Min.	:190.5
1st Qu.:	2000	1st Qu.:	17.79	1st Qu.:	9.257	1st Qu.:	347.7
Median	:2008	Median	:18.54	Median	: 9.944	Median	:461.1
Mean	:2008	Mean	:18.51	Mean	: 9.824	Mean	:441.9
3rd Qu.:	2016	3rd Qu.:	19.18	3rd Qu.:	10.277	3rd Qu.:	567.2
Max.	:2024	Max.	:20.40	Max.	:11.401	Max.	:807.3

rain_days_annual	mean_wind_annual	lag_mean_temp_annual	lag_total_rain_annual
Min. : 39.00	Min. :2.951	Min. :17.01	Min. :190.5
1st Qu.: 65.00	1st Qu.:3.381	1st Qu.:17.78	1st Qu.:346.9
Median : 76.00	Median :3.714	Median :18.53	Median :463.5
Mean : 74.18	Mean :3.670	Mean :18.49	Mean :444.9
3rd Qu.: 84.00	3rd Qu.:3.876	3rd Qu.:19.06	3rd Qu.:567.5
Max. :107.00	Max. :4.537	Max. :20.40	Max. :807.3
		NA's :1	NA's :1

```
lag_rain_days_annual
Min. : 39.00
1st Qu.: 65.75
Median : 77.00
Mean : 74.69
3rd Qu.: 84.25
Max. :107.00
NA's :1
```

```
timing_df <- vultures.clean |>
  dplyr::filter(!is.na(laying.date)) |>
  dplyr::mutate(
    year = lubridate::year(laying.date),
    lay_DOY = lubridate::yday(laying.date)
  )
```

B.1.9 Creating 30-day window data

```
timing_by_year <- timing_df |>
  dplyr::mutate(year = as.integer(year)) |>
  dplyr::filter(!is.na(year), year != 1900) |>
  dplyr::group_by(year) |>
  dplyr::summarise(
    lay_DOY_median = median(lay_DOY, na.rm = TRUE),
    lay_DOY_mean = mean(lay_DOY, na.rm = TRUE),
    n_nests = sum(!is.na(lay_DOY)),
    first_DOY = min(lay_DOY, na.rm = TRUE),
    last_DOY = max(lay_DOY, na.rm = TRUE),
    .groups = "drop"
  ) |>
  dplyr::arrange(year)

timing_by_year
```

A tibble: 32 x 6

	year	lay_DOY_median	lay_DOY_mean	n_nests	first_DOY	last_DOY
	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>
1	1993	148.	148.	28	103	203
2	1994	159	162.	39	128	212
3	1995	156.	154.	22	132	171

```

4 1996      158      158.    35    129    209
5 1997      151      152.    46    132    190
6 1998      148      151.    45    122    188
7 1999      152      153.    37    136    192
8 2000      153      155.    41    124    201
9 2001      156.      158.    56    129    208
10 2002      154.      155.    30     20    185
# i 22 more rows

```

```
nrow(timing_by_year)
```

```
[1] 32
```

```
n_distinct(timing_by_year$year)
```

```
[1] 32
```

```
summary(timing_by_year$n_nests)
```

```

      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
22.00   37.75   45.50   45.59   53.00   67.00

```

```

obj3_year_df <- timing_by_year |>
  left_join(weather_march_by_year, by = "year") |>
  arrange(year)

obj3_year_df

```

```

# A tibble: 32 x 16
   year lay_DOY_median lay_DOY_mean n_nests first_DOY last_DOY mean_temp_win
  <dbl>      <dbl>      <dbl>   <int>    <dbl>    <dbl>      <dbl>
1 1993      148.      148.    28     103     203      21.0
2 1994      159      162.    39     128     212      21.7
3 1995      156.      154.    22     132     171      22.1
4 1996      158      158.    35     129     209      22.3
5 1997      151      152.    46     132     190      19.8
6 1998      148      151.    45     122     188      22.2
7 1999      152      153.    37     136     192      25.4
8 2000      153      155.    41     124     201      22.3
9 2001      156.      158.    56     129     208      23.2
10 2002      154.      155.    30      20     185      23.5
# i 22 more rows
# i 9 more variables: max_temp_win <dbl>, min_temp_win <dbl>,
#   total_rain_win <dbl>, max_rain_win <dbl>, rain_days_win <int>,
#   mean_wind_win <dbl>, max_wind_win <dbl>, hail_days_win <dbl>, n_days <int>

```

B.1.10 Annual weather summaries

```

weather_yearly <- WeatherData_clean |>
  group_by(YEAR) |>
  summarise(
    mean_temp = mean(TEMP, na.rm = TRUE),
    max_temp = max(MAX, na.rm = TRUE),
    min_temp = min(MIN, na.rm = TRUE),
    total_rain = sum(PRCP, na.rm = TRUE),
    max_rain = max(PRCP, na.rm = TRUE),
    rain_days = sum(PRCP > 0, na.rm = TRUE),
    mean_wind = mean(WDSP, na.rm = TRUE),
    max_wind = max(MXSPD, na.rm = TRUE),
    hail_days = sum(I_HAIL, na.rm = TRUE),
    .groups = "drop"
  )

weather_yearly |>
  filter(YEAR == 2004) |>
  select(YEAR, total_rain, max_rain, rain_days)

```

```

# A tibble: 1 x 4
  YEAR total_rain max_rain rain_days
<int>    <dbl>    <dbl>    <int>
1  2004      461.     28.8      54

```

B.1.11 Weather trends over time

```

# Temperature trends
summary(lm(mean_temp ~ YEAR, data = weather_yearly))

```

```

Call:
lm(formula = mean_temp ~ YEAR, data = weather_yearly)

```

```

Residuals:
    Min       1Q   Median       3Q      Max
-1.49859 -0.73263  0.01794  0.68011  1.91081

```

```

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  23.67507   31.36043   0.755   0.456
YEAR         -0.00257    0.01562  -0.165   0.870

```

```

Residual standard error: 0.8543 on 31 degrees of freedom
Multiple R-squared:  0.0008729, Adjusted R-squared:  -0.03136
F-statistic: 0.02708 on 1 and 31 DF,  p-value: 0.8704

```

```

summary(lm(max_temp ~ YEAR, data = weather_yearly))

```

```

Call:
lm(formula = max_temp ~ YEAR, data = weather_yearly)

```

```

Residuals:

```

	Min	1Q	Median	3Q	Max
	-3.4194	-1.4481	-0.1481	0.7663	5.1377

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-103.42807	74.27533	-1.392	0.1737
YEAR	0.07142	0.03699	1.931	0.0627

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.023 on 31 degrees of freedom

Multiple R-squared: 0.1074, Adjusted R-squared: 0.07857

F-statistic: 3.728 on 1 and 31 DF, p-value: 0.06268

```
summary(lm(min_temp ~ YEAR, data = weather_yearly))
```

Call:

```
lm(formula = min_temp ~ YEAR, data = weather_yearly)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.6621	-1.1999	0.0613	0.8673	3.2483

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-32.37852	55.87587	-0.579	0.566
YEAR	0.01293	0.02783	0.465	0.645

Residual standard error: 1.522 on 31 degrees of freedom

Multiple R-squared: 0.006922, Adjusted R-squared: -0.02511

F-statistic: 0.2161 on 1 and 31 DF, p-value: 0.6453

```
# Rainfall trends
```

```
summary(lm(total_rain ~ YEAR, data = weather_yearly))
```

Call:

```
lm(formula = total_rain ~ YEAR, data = weather_yearly)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-296.19	-95.76	8.01	86.31	351.34

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6056.800	5445.968	1.112	0.275
YEAR	-2.796	2.712	-1.031	0.311

Residual standard error: 148.3 on 31 degrees of freedom

Multiple R-squared: 0.03315, Adjusted R-squared: 0.001965

F-statistic: 1.063 on 1 and 31 DF, p-value: 0.3105

```
summary(lm(rain_days ~ YEAR, data = weather_yearly))
```

Call:

```
lm(formula = rain_days ~ YEAR, data = weather_yearly)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-32.538	-9.126	-0.448	11.707	32.252

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-305.0027	613.8635	-0.497	0.623
YEAR	0.1888	0.3057	0.618	0.541

Residual standard error: 16.72 on 31 degrees of freedom

Multiple R-squared: 0.01216, Adjusted R-squared: -0.01971

F-statistic: 0.3816 on 1 and 31 DF, p-value: 0.5413

```
summary(lm(max_rain ~ YEAR, data = weather_yearly))
```

Call:

```
lm(formula = max_rain ~ YEAR, data = weather_yearly)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-25.448	-15.957	-6.285	14.454	60.935

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	592.4045	792.3973	0.748	0.460
YEAR	-0.2721	0.3946	-0.690	0.496

Residual standard error: 21.59 on 31 degrees of freedom

Multiple R-squared: 0.01511, Adjusted R-squared: -0.01666

F-statistic: 0.4755 on 1 and 31 DF, p-value: 0.4956

```
# Wind trends
summary(lm(mean_wind ~ YEAR, data = weather_yearly))
```

Call:

```
lm(formula = mean_wind ~ YEAR, data = weather_yearly)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.73374	-0.18419	-0.03271	0.14879	0.92830

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
--	----------	------------	---------	----------

```
(Intercept) 34.014073 14.135469 2.406 0.0223 *
YEAR -0.015112 0.007039 -2.147 0.0398 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.3851 on 31 degrees of freedom
Multiple R-squared: 0.1294, Adjusted R-squared: 0.1013
F-statistic: 4.608 on 1 and 31 DF, p-value: 0.03975
```

```
summary(lm(max_wind ~ YEAR, data = weather_yearly))
```

```
Call:
lm(formula = max_wind ~ YEAR, data = weather_yearly)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-6.003 -3.491 -1.363  2.440  9.249
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 194.80838   176.22563    1.105   0.277
YEAR        -0.08790    0.08776   -1.002   0.324
```

```
Residual standard error: 4.8 on 31 degrees of freedom
Multiple R-squared: 0.03135, Adjusted R-squared: 9.998e-05
F-statistic: 1.003 on 1 and 31 DF, p-value: 0.3243
```

B.1.12 Lagged weather summaries

```
weather_yearly_lag <- weather_yearly |>
  arrange(YEAR) |>
  mutate(across(
    c(mean_temp, max_temp, min_temp, total_rain, max_rain, rain_days,
      mean_wind, max_wind, hail_days),
    ~ lag(.x, 1),
    .names = "lag_{.col}"
  ))
```

B.1.13 Correlations between weather variables

```
vars <- weather_yearly |>
  select(mean_temp, max_temp, min_temp,
         total_rain, max_rain, rain_days,
         mean_wind, max_wind, hail_days)

res <- rcorr(as.matrix(vars), type = "pearson")

cor_df <- as.data.frame(res$r) |>
  rownames_to_column("var1") |>
  pivot_longer(~var1, names_to = "var2", values_to = "cor") |>
  mutate(p = as.vector(res$p))

cor_df
```



```
# A tibble: 81 x 4
  var1      var2      cor      p
  <chr>    <chr>    <dbl>   <dbl>
1 mean_temp mean_temp 1      NA
2 mean_temp max_temp  0.546  0.00102
3 mean_temp min_temp  0.264  0.138
4 mean_temp total_rain -0.216  0.227
5 mean_temp max_rain  0.109  0.546
6 mean_temp rain_days -0.444  0.00961
7 mean_temp mean_wind -0.142  0.431
8 mean_temp max_wind  0.138  0.443
9 mean_temp hail_days  0.172  0.337
10 max_temp mean_temp  0.546  0.00102
# i 71 more rows
```

B.1.14 Trends over time for breeding variables

B.1.15 Number of active nests

```
summary(lm(active_nests ~ year, data = active_nests_yearly))
```

Call:

```
lm(formula = active_nests ~ year, data = active_nests_yearly)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-26.189	-6.508	4.153	7.167	20.079

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4568.5150	413.0744	-11.06	4.17e-12 ***
year	2.3154	0.2057	11.26	2.70e-12 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.74 on 30 degrees of freedom

Multiple R-squared: 0.8086, Adjusted R-squared: 0.8022

F-statistic: 126.8 on 1 and 30 DF, p-value: 2.699e-12

B.1.16 Breeding success

```
success_year <- glm(
  cbind(success_nests, failed_nests) ~ year,
  family = binomial,
  data = success_yearly
)

summary(success_year)
```

```
Call:
glm(formula = cbind(success_nests, failed_nests) ~ year, family = binomial,
     data = success_yearly)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	45.434641	8.858715	5.129	2.92e-07 ***
year	-0.022505	0.004405	-5.109	3.24e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 107.689 on 31 degrees of freedom
 Residual deviance: 81.283 on 30 degrees of freedom
 AIC: 237.58

Number of Fisher Scoring iterations: 3

```
check_overdispersion(success_year)
```

```
[1] 2.670817
```

B.1.17 Breeding timing of succesful nests

```
timing_yearly <- vultures.clean |>
  mutate(
    year = year(ringing.date),
    lay_DOY = yday(laying.date)
  ) |>
  filter(!is.na(year), !is.na(lay_DOY)) |>
  group_by(year) |>
  summarise(
    mean_lay_DOY = mean(lay_DOY, na.rm = TRUE),
    .groups = "drop"
  )

timing_yearly_clean <- timing_yearly |>
  filter(year >= 1992)

summary(lm(mean_lay_DOY ~ year, data = timing_yearly_clean))
```

Call:

```
lm(formula = mean_lay_DOY ~ year, data = timing_yearly_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.3676	-2.6114	-0.0087	3.5078	7.7270

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	312.43769	165.46254	1.888	0.0687 .

```

year          -0.07915    0.08238  -0.961    0.3443
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.303 on 30 degrees of freedom
Multiple R-squared:  0.02985,    Adjusted R-squared:  -0.002487
F-statistic: 0.9231 on 1 and 30 DF,  p-value: 0.3443

```

B.2 Objective 1: Number of active nests

B.2.1 DAGS

B.2.2 Scenario A

```

dir.create("figures", showWarnings = FALSE)

export_dag_pdf <- function(dag, filename, w = 8, h = 4) {
  svg_txt <- DiagrammeRsvg::export_svg(dag)
  rsvg::rsvg_pdf(
    charToRaw(svg_txt),
    file = file.path("figures", filename),
    width = w, height = h
  )
  file.info(file.path("figures", filename))[, "size"]
}

trend_confounded_dag <- DiagrammeR::grViz("
digraph trend_confounded {
  graph [
    layout = neato,
    splines = false,
    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    shape = box,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]

  Y [label = 'Y', pos = '2.5,2!']
  W [label = 'W', pos = '1.2,0!']
  N [label = 'N', pos = '4.0,0!']

  Y -> W
  Y -> N
  W -> N
}
")

export_dag_pdf(trend_confounded_dag, "trend_confounded_dag.pdf", 8, 4)

```

B.2.3 Scenario B

```
trend_W_B_dag <- DiagrammeR::grViz("
digraph trend_W_B {
  graph [
    layout = neato,
    splines = false,
    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    shape = box,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]

  Y [label = 'Y', pos = '0,0!']
  W [label = 'W', pos = '3,0!']
  N [label = 'N', pos = '6,0!']

  Y -> W
  W -> N
}
")

export_dag_pdf(trend_W_B_dag, "trend_W_B_dag.pdf", 8, 4)
```

[1] 5915

B.2.4 Scenario C

```
trend_mediator_dag <- DiagrammeR::grViz("
digraph trend_mediator {
  graph [
    layout = neato,
    splines = false,
    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    shape = box,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]

  W [label = 'W', pos = '0,0!']
  Y [label = 'Y', pos = '3,0!']
```

```

N [label = 'N', pos = '6,0!']

W -> Y
Y -> N
}
")

export_dag_pdf(trend_mediator_dag, "trend_mediator_dag.pdf", 8, 4)

```

[1] 5910

B.2.5 Models

```

objective1_table <- tribble(
  ~`Hypothesis / justification`, ~Model,

  "**Temp**", NA,
  "Warmer years increase energetic cost of adult birds, reducing likelihood to breed.",
  "active_nests ~ mean_temp",
  "Extreme heat events cause thermal stress, reducing breeding effort (active cooling is energetically costly and may
  ⇨ reduce adult condition).",
  "active_nests ~ max_temp",
  "Cold extremes increase thermoregulatory demand (often increasing time spent foraging), reducing effort put into
  ⇨ breeding",
  "active_nests ~ min_temp",

  "**Rain**", NA,
  "High rainfall reduces foraging efficiency and adult condition, reducing breeding effort.",
  "active_nests ~ total_rain",
  "Prolonged rainfall conditions reduce breeding effort by limiting foraging opportunity (fewer or weaker thermals
  ⇨ for soaring)",
  "active_nests ~ rain_days",
  "Intense rainfall events decrease foraging efficiency and nest attendance, reducing breeding effort.",
  "active_nests ~ max_rain",
  "Previous year's rainfall influences food availability and adult condition, affecting breeding effort in the
  ⇨ following year.",
  "active_nests ~ lag_total_rain",

  "**Wind**", NA,
  "Persistent windy conditions increase the cost of flying, limiting foraging and the energy available for
  ⇨ reproduction, reducing breeding effort",
  "active_nests ~ mean_wind",
  "Extreme winds increase nest damage risk, reducing the likelihood that adults will attempt breeding (WBV often
  ⇨ reuse nests from previous years) ",
  "active_nests ~ max_wind",

  "**Hail**", NA,
  "Severe storm events increase disturbance and nest damage risk, reducing breeding effort",
  "active_nests ~ hail_days",

  "**Joint effects**", NA,
  "Cold stress combined with prolonged wet conditions jointly limit breeding effort",
  "active_nests ~ min_temp + total_rain",
  "Cold stress combined with intense rainfall events reduces breeding effort",
  "active_nests ~ min_temp + max_rain",
  "Cold extremes combined with extreme wind increase energetic costs and breeding risk, reducing breeding effort",
  "active_nests ~ min_temp + max_wind",
  "Severe storm exposure combined with extreme wind increases nest disturbance, reducing breeding effort",
  "active_nests ~ hail_days + max_wind",
  "Reduced thermals for soaring combined with high flight costs limit breeding effort",
  "active_nests ~ max_rain + max_wind",
  "Heat stress combined with prolonged wet conditions jointly reduce breeding effort by increasing energetic costs
  ⇨ while limiting foraging",

```

```

"active_nests ~ max_temp + rain_days",
"High annual rainfall and frequent hail events each impose energetic and disturbance costs that reduce breeding
↪ effort.",
"active_nests ~ hail_days + rain_days",

"**Interactive effects**", NA,
"The negative effect of cold extremes on breeding effort is increased in persistently wet years.",
"active_nests ~ min_temp * rain_days",
"Cold-related energetic stress is amplified in windy years due to increased heat loss and flight costs",
"active_nests ~ min_temp * mean_wind",
"Adult condition resulting from the previous year interacts with current-year heat stress to influence breeding
↪ effort",
"active_nests ~ lag_total_rain * max_temp"
) |>
mutate(Model = if_else(is.na(Model), "", Model))

knitr::kable(
  objective1_table,
  align = c("l", "l"),
  caption = "Objective 1 candidate model set for breeding effort (active nests)"
)

```

Table 2: Objective 1 candidate model set for breeding effort (active nests)

Hypothesis / justification	Model
Temp	
Warmer years increase energetic cost of adult birds, reducing likelihood to breed.	active_nests ~ mean_temp
Extreme heat events cause thermal stress, reducing breeding effort (active cooling is energetically costly and may reduce adult condition).	active_nests ~ max_temp
Cold extremes increase thermoregulatory demand (often increasing time spent foraging), reducing effort put into breeding	active_nests ~ min_temp
Rain	
High rainfall reduces foraging efficiency and adult condition, reducing breeding effort.	active_nests ~ total_rain
Prolonged rainfall conditions reduce breeding effort by limiting foraging opportunity (fewer or weaker thermals for soaring)	active_nests ~ rain_days
Intense rainfall events decrease foraging efficiency and nest attendance, reducing breeding effort.	active_nests ~ max_rain
Previous year's rainfall influences food availability and adult condition, affecting breeding effort in the following year.	active_nests ~ lag_total_rain
Wind	
Persistent windy conditions increase the cost of flying, limiting foraging and the energy available for reproduction, reducing breeding effort	active_nests ~ mean_wind
Extreme winds increase nest damage risk, reducing the likelihood that adults will attempt breeding (WBV often reuse nests from previous years)	active_nests ~ max_wind
Hail	
Severe storm events increase disturbance and nest damage risk, reducing breeding effort	active_nests ~ hail_days
Joint effects	
Cold stress combined with prolonged wet conditions jointly limit breeding effort	active_nests ~ min_temp + total_rain
Cold stress combined with intense rainfall events reduces breeding effort	active_nests ~ min_temp + max_rain
Cold extremes combined with extreme wind increase energetic costs and breeding risk, reducing breeding effort	active_nests ~ min_temp + max_wind
Severe storm exposure combined with extreme wind increases nest disturbance, reducing breeding effort	active_nests ~ hail_days + max_wind
Reduced thermals for soaring combined with high flight costs limit breeding effort	active_nests ~ max_rain + max_wind

Hypothesis / justification	Model
Heat stress combined with prolonged wet conditions jointly reduce breeding effort by increasing energetic costs while limiting foraging	active_nests ~ max_temp + rain_days
High annual rainfall and frequent hail events each impose energetic and disturbance costs that reduce breeding effort.	active_nests ~ hail_days + rain_days
Interactive effects	
The negative effect of cold extremes on breeding effort is increased in persistently wet years.	active_nests ~ min_temp * rain_days
Cold-related energetic stress is amplified in windy years due to increased heat loss and flight costs	active_nests ~ min_temp * mean_wind
Adult condition resulting from the previous year interacts with current-year heat stress to influence breeding effort	active_nests ~ lag_total_rain * max_temp

B.2.6 Data for number of active nest models

```
# number of active nests per year
active.nests <- vultures.clean |>
  mutate(year = year(ringing.date)) |>
  group_by(year) |>
  summarise(active_nests = n(), .groups = "drop")

# Join to weather, create log effort response
effort_weather <- active.nests |>
  left_join(weather_yearly_lag, by = c("year" = "YEAR")) |>
  filter(!is.na(lag_total_rain)) |>
  mutate(
    log_active_nests = log(active_nests)
  )
```

B.2.7 Fitting the no-year models

```
# Baseline
m_null <- lm(log_active_nests ~ 1, data = effort_weather)

# Single predictor models
m_mean_temp <- lm(log_active_nests ~ mean_temp, data = effort_weather)
m_max_temp <- lm(log_active_nests ~ max_temp, data = effort_weather)
m_min_temp <- lm(log_active_nests ~ min_temp, data = effort_weather)

m_total_rain <- lm(log_active_nests ~ total_rain, data = effort_weather)
m_rain_days <- lm(log_active_nests ~ rain_days, data = effort_weather)
m_max_rain <- lm(log_active_nests ~ max_rain, data = effort_weather)
m_lag_total_rain <- lm(log_active_nests ~ lag_total_rain, data = effort_weather)

m_mean_wind <- lm(log_active_nests ~ mean_wind, data = effort_weather)
m_max_wind <- lm(log_active_nests ~ max_wind, data = effort_weather)
m_hail_days <- lm(log_active_nests ~ hail_days, data = effort_weather)

# Joint effects
m_minT_totalR <- lm(log_active_nests ~ min_temp + total_rain, data = effort_weather)
m_minT_maxR <- lm(log_active_nests ~ min_temp + max_rain, data = effort_weather)
m_minT_maxW <- lm(log_active_nests ~ min_temp + max_wind, data = effort_weather)
m_hail_maxW <- lm(log_active_nests ~ hail_days + max_wind, data = effort_weather)
m_maxR_maxW <- lm(log_active_nests ~ max_rain + max_wind, data = effort_weather)
m_maxT_rainDays <- lm(log_active_nests ~ max_temp + rain_days, data = effort_weather)
m_hail_rainDays <- lm(log_active_nests ~ hail_days + rain_days, data = effort_weather)
```

```
# Interactions
m_minT_x_rainDays <- lm(log_active_nests ~ min_temp * rain_days,      data = effort_weather)
m_minT_x_meanWind <- lm(log_active_nests ~ min_temp * mean_wind,    data = effort_weather)
m_lagR_x_maxT     <- lm(log_active_nests ~ lag_total_rain * max_temp, data = effort_weather)
```

B.2.8 Fit with year models

```
# Baseline
m_year <- lm(log_active_nests ~ year, data = effort_weather)

# Single predictor models
m_mean_temp_year <- lm(log_active_nests ~ year + mean_temp,      data = effort_weather)
m_max_temp_year <- lm(log_active_nests ~ year + max_temp,       data = effort_weather)
m_min_temp_year <- lm(log_active_nests ~ year + min_temp,       data = effort_weather)

m_total_rain_year <- lm(log_active_nests ~ year + total_rain,    data = effort_weather)
m_rain_days_year <- lm(log_active_nests ~ year + rain_days,     data = effort_weather)
m_max_rain_year <- lm(log_active_nests ~ year + max_rain,       data = effort_weather)
m_lag_total_rain_year <- lm(log_active_nests ~ year + lag_total_rain, data = effort_weather)

m_mean_wind_year <- lm(log_active_nests ~ year + mean_wind,    data = effort_weather)
m_max_wind_year <- lm(log_active_nests ~ year + max_wind,      data = effort_weather)
m_hail_days_year <- lm(log_active_nests ~ year + hail_days,     data = effort_weather)

# Joint effects
m_minT_totalR_year <- lm(log_active_nests ~ year + min_temp + total_rain, data = effort_weather)
m_minT_maxR_year <- lm(log_active_nests ~ year + min_temp + max_rain, data = effort_weather)
m_minT_maxW_year <- lm(log_active_nests ~ year + min_temp + max_wind, data = effort_weather)
m_hail_maxW_year <- lm(log_active_nests ~ year + hail_days + max_wind, data = effort_weather)
m_maxR_maxW_year <- lm(log_active_nests ~ year + max_rain + max_wind, data = effort_weather)
m_maxT_rainDays_year <- lm(log_active_nests ~ year + max_temp + rain_days, data = effort_weather)
m_hail_rainDays_year <- lm(log_active_nests ~ year + hail_days + rain_days, data = effort_weather)

# Interactions
m_minT_x_rainDays_year <- lm(log_active_nests ~ year + min_temp * rain_days,      data = effort_weather)
m_minT_x_meanWind_year <- lm(log_active_nests ~ year + min_temp * mean_wind,    data = effort_weather)
m_lagR_x_maxT_year <- lm(log_active_nests ~ year + lag_total_rain * max_temp, data = effort_weather)
```

B.2.9 Two separate AICc model selection runs

```
## AIC 1: without year
cand_effort_no_year <- list(
  null = m_null,

  mean_temp = m_mean_temp,
  max_temp = m_max_temp,
  min_temp = m_min_temp,
  total_rain = m_total_rain,
  rain_days = m_rain_days,
  max_rain = m_max_rain,
  lag_total_rain = m_lag_total_rain,
  mean_wind = m_mean_wind,
  max_wind = m_max_wind,
  hail_days = m_hail_days,

  minT_totalR = m_minT_totalR,
  minT_maxR = m_minT_maxR,
  minT_maxW = m_minT_maxW,
  hail_maxW = m_hail_maxW,
```



```

maxR_maxW      = m_maxR_maxW,
maxT_rainDays  = m_maxT_rainDays,
hail_rainDays  = m_hail_rainDays,

minT_x_rainDays = m_minT_x_rainDays,
minT_x_meanWind = m_minT_x_meanWind,
lagR_x_maxT     = m_lagR_x_maxT
)

aic_effort_no_year <- aictab(cand.set = cand_effort_no_year, modnames = names(cand_effort_no_year))
aic_effort_no_year

```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
total_rain	3	17.13	0.00	0.14	0.14	-5.14
max_rain	3	17.45	0.32	0.12	0.27	-5.30
max_temp	3	17.80	0.67	0.10	0.37	-5.47
hail_days	3	17.96	0.83	0.10	0.46	-5.55
maxR_maxW	4	18.03	0.90	0.09	0.56	-4.27
maxT_rainDays	4	18.78	1.65	0.06	0.62	-4.65
mean_wind	3	19.00	1.86	0.06	0.68	-6.07
null	2	19.44	2.31	0.05	0.72	-7.51
minT_maxR	4	19.66	2.53	0.04	0.76	-5.09
minT_totalR	4	19.73	2.60	0.04	0.80	-5.12
hail_maxW	4	19.90	2.77	0.04	0.84	-5.21
hail_rainDays	4	20.11	2.98	0.03	0.87	-5.32
lagR_x_maxT	5	20.18	3.04	0.03	0.90	-3.93
max_wind	3	20.61	3.48	0.03	0.93	-6.88
min_temp	3	21.38	4.25	0.02	0.94	-7.26
mean_temp	3	21.61	4.48	0.02	0.96	-7.38
lag_total_rain	3	21.73	4.60	0.01	0.97	-7.43
rain_days	3	21.85	4.72	0.01	0.99	-7.50
minT_maxW	4	23.01	5.88	0.01	0.99	-6.76
minT_x_meanWind	5	24.01	6.88	0.00	1.00	-5.85
minT_x_rainDays	5	26.33	9.19	0.00	1.00	-7.01

```

## AIC 2: with year
cand_effort_with_year <- list(
  year          = m_year,

  mean_temp     = m_mean_temp_year,
  max_temp      = m_max_temp_year,
  min_temp      = m_min_temp_year,
  total_rain    = m_total_rain_year,
  rain_days     = m_rain_days_year,
  max_rain      = m_max_rain_year,
  lag_total_rain = m_lag_total_rain_year,
  mean_wind     = m_mean_wind_year,
  max_wind      = m_max_wind_year,
  hail_days     = m_hail_days_year,

  minT_totalR   = m_minT_totalR_year,
  minT_maxR     = m_minT_maxR_year,
  minT_maxW     = m_minT_maxW_year,
  hail_maxW     = m_hail_maxW_year,
  maxR_maxW     = m_maxR_maxW_year,
  maxT_rainDays = m_maxT_rainDays_year,

```

```

hail_rainDays      = m_hail_rainDays_year,

minT_x_rainDays    = m_minT_x_rainDays_year,
minT_x_meanWind    = m_minT_x_meanWind_year,
lagR_x_maxT        = m_lagR_x_maxT_year
)

aic_effort_with_year <- aictab(cand.set = cand_effort_with_year, modnames = names(cand_effort_with_year))
aic_effort_with_year

```

Model selection based on AICc:

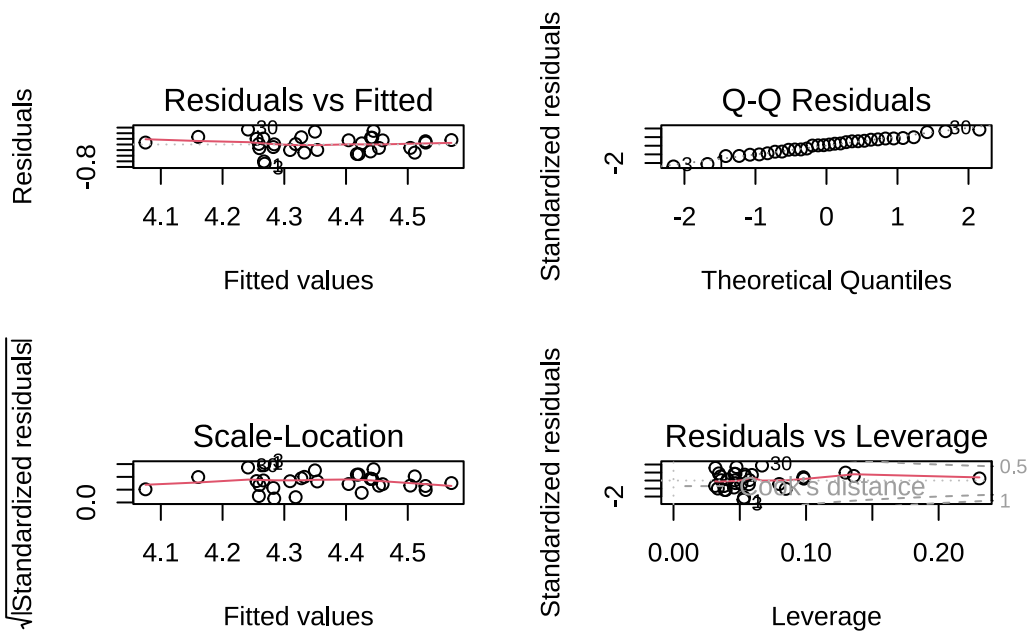
	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
max_rain	4	-30.85	0.00	0.39	0.39	20.17
minT_maxR	5	-29.55	1.31	0.20	0.59	20.93
maxR_maxW	5	-28.14	2.71	0.10	0.69	20.22
year	3	-27.09	3.76	0.06	0.75	16.97
total_rain	4	-26.37	4.49	0.04	0.79	17.92
min_temp	4	-25.97	4.88	0.03	0.82	17.73
mean_temp	4	-25.27	5.58	0.02	0.84	17.38
lag_total_rain	4	-25.17	5.68	0.02	0.87	17.32
max_temp	4	-24.99	5.86	0.02	0.89	17.24
mean_wind	4	-24.70	6.16	0.02	0.91	17.09
rain_days	4	-24.50	6.35	0.02	0.92	16.99
hail_days	4	-24.47	6.38	0.02	0.94	16.98
max_wind	4	-24.47	6.38	0.02	0.95	16.97
minT_totalR	5	-24.37	6.49	0.02	0.97	18.34
minT_maxW	5	-23.25	7.60	0.01	0.98	17.78
maxT_rainDays	5	-22.55	8.30	0.01	0.98	17.43
hail_rainDays	5	-21.69	9.16	0.00	0.99	17.00
hail_maxW	5	-21.65	9.21	0.00	0.99	16.98
lagR_x_maxT	6	-21.46	9.39	0.00	0.99	18.41
minT_x_meanWind	6	-21.16	9.69	0.00	1.00	18.26
minT_x_rainDays	6	-20.32	10.53	0.00	1.00	17.84

```

## Best model (no year)
best_no_year_name <- aic_effort_no_year$Modnames[1]
best_no_year_model <- cand_effort_no_year[[best_no_year_name]]

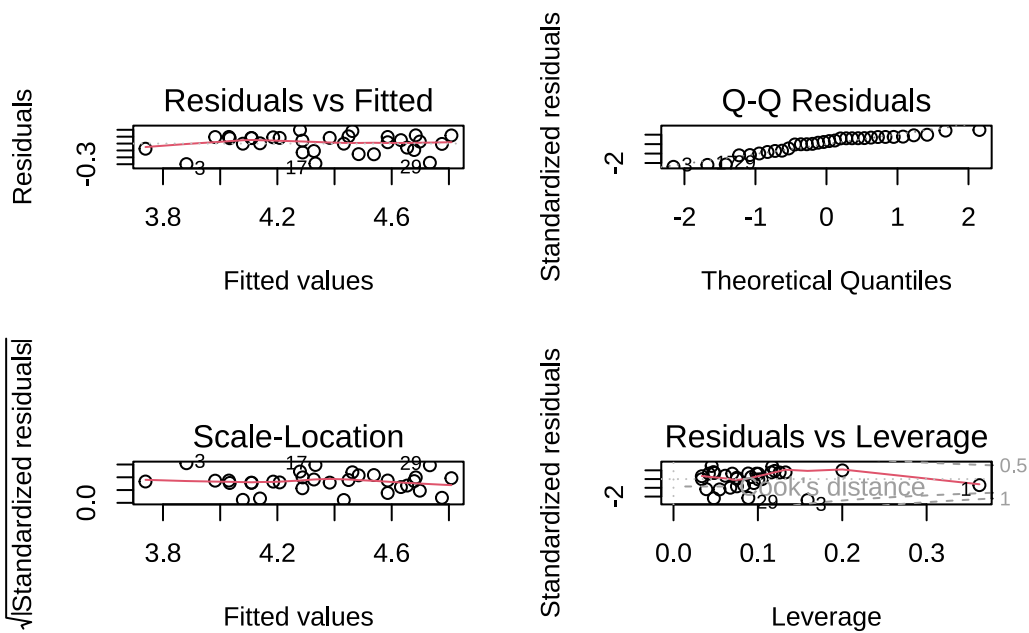
par(mfrow = c(2,2))
plot(best_no_year_model)

```



```
## Best model (with year)
best_with_year_name <- aic_effort_with_year$Modnames[1]
best_with_year_model <- cand_effort_with_year[[best_with_year_name]]

par(mfrow = c(2,2))
plot(best_with_year_model)
```



```
## Adjusted R2 summaries (as you had)
summary(m_year)$adj.r.squared
```

```
[1] 0.776371
```

```
summary(best_with_year_model)$adj.r.squared
```

```
[1] 0.8105132
```

```
summary(best_no_year_model)$adj.r.squared
```

```
[1] 0.1093628
```

B.2.10 Number of active nests over time

```
ggplot(effort_weather, aes(x = year, y = active_nests)) +  
  geom_point(  
    size = 2.4,  
    colour = "black",  
    alpha = 0.9  
  ) +  
  geom_smooth(  
    method = "lm",  
    se = TRUE,  
    colour = "#D5D6D2",  
    fill = "#D5D6D2",  
    alpha = 0.25,  
    linewidth = 1.8  
  ) +  
  scale_x_continuous(  
    breaks = seq(  
      min(effort_weather$year),  
      max(effort_weather$year),  
      by = 5  
    )  
  ) +  
  labs(  
    x = "Year",  
    y = "Number of active nests"  
  ) +  
  theme_bw(base_family = "Times New Roman") +  
  theme(  
    panel.grid = element_blank(),  
    axis.title = element_text(face = "bold", size = 12),  
    axis.text = element_text(face = "bold", size = 10)  
  )
```

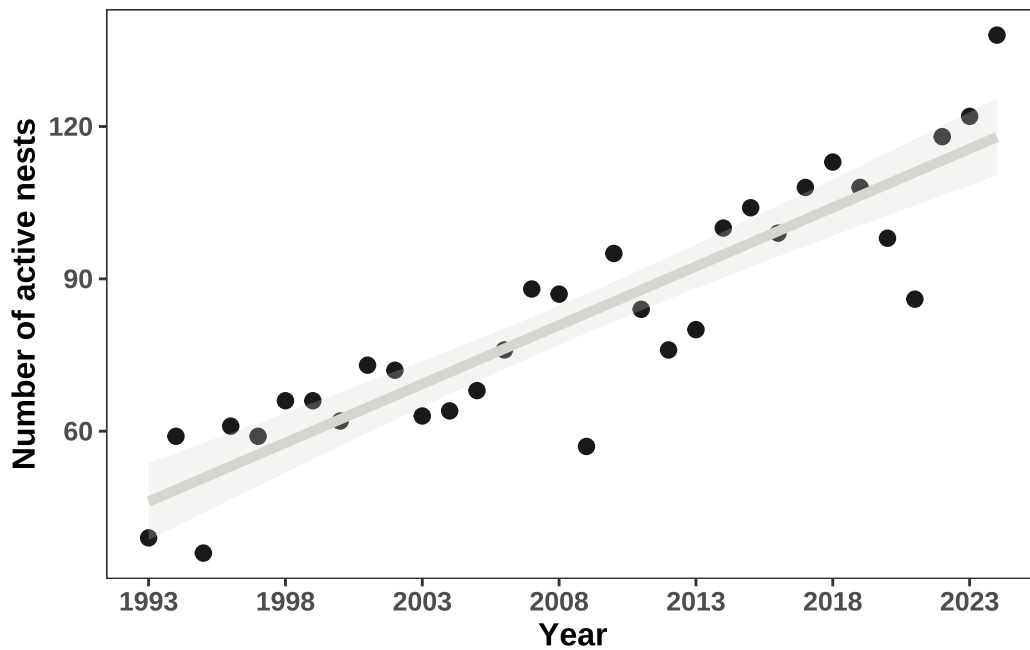


Figure 4: The number of active nests at Dronfield, over time (1993-2024)

B.2.11 AIC tables: with and without year

```
library(dplyr)
library(tibble)
library(officer)
library(flextable)

if (requireNamespace("conflicted", quietly = TRUE)) {
  conflicted::conflicts_prefer(flextable::compose)
}

stopifnot(exists("aic_effort_with_year"))
stopifnot(exists("cand_effort_with_year"))
stopifnot(exists("aic_effort_no_year"))
stopifnot(exists("cand_effort_no_year"))

group_order <- c("Baseline", "Rain", "Temperature", "Wind", "Hail", "Joint effects", "Interactions")

model_group <- function(m){
  dplyr::case_when(
    m %in% c("null", "year") ~ "Baseline",
    m %in% c("total_rain", "rain_days", "max_rain", "lag_total_rain") ~ "Rain",
    m %in% c("mean_temp", "max_temp", "min_temp") ~ "Temperature",
    m %in% c("mean_wind", "max_wind") ~ "Wind",
    m %in% c("hail_days") ~ "Hail",
    m %in% c("minT_totalR", "minT_maxR", "minT_maxW", "hail_maxW", "maxR_maxW", "maxT_rainDays", "hail_rainDays") ~ "Joint
  effects",
    m %in% c("minT_x_rainDays", "minT_x_meanWind", "lagR_x_maxT") ~ "Interactions",
    TRUE ~ "Joint effects"
  )
}

label_model <- function(x){
  dplyr::recode(
    x,
    "null" = "Intercept only",
    "year" = "Year (baseline)",
  )
}
```

```

"total_rain"      = "Total rainfall",
"rain_days"      = "Rainy days",
"max_rain"       = "Maximum daily rainfall",
"lag_total_rain" = "Previous year's total rainfall",
"mean_temp"      = "Mean temperature",
"max_temp"       = "Maximum temperature",
"min_temp"       = "Minimum temperature",
"mean_wind"      = "Mean wind speed",
"max_wind"       = "Maximum wind speed",
"hail_days"      = "Hail days",
"minT_totalR"    = "Min temperature + total rainfall",
"minT_maxR"      = "Min temperature + max daily rainfall",
"minT_maxW"      = "Min temperature + max wind speed",
"hail_maxW"      = "Hail days + max wind speed",
"maxR_maxW"      = "Max daily rainfall + max wind speed",
"maxT_rainDays"  = "Max temperature + rainy days",
"hail_rainDays"  = "Hail days + rainy days",
"minT_x_rainDays" = "Min temperature × rainy days",
"minT_x_meanWind" = "Min temperature × mean wind speed",
"lagR_x_maxT"    = "Prev. rainfall × max temperature",
.default = x
)
}

single_weather_models <- c(
  "total_rain", "rain_days", "max_rain", "lag_total_rain",
  "mean_temp", "max_temp", "min_temp",
  "mean_wind", "max_wind",
  "hail_days"
)

get_slope_se_single_weather <- function(mod, model_id){
  slope_se <- ""
  if (model_id %in% single_weather_models) {
    cn <- names(coef(mod))
    cn <- setdiff(cn, "(Intercept)")
    cn_weather <- cn[!grepl("^year$|^year\\b|\\byear\\b", cn)]
    if (length(cn_weather) == 1) {
      term <- cn_weather[1]
      est <- coef(mod)[term]
      se <- sqrt(diag(vcov(mod)))[term]
      slope_se <- paste0(round(est, 3), " (", round(se, 3), ")")
    }
  }
  slope_se
}

prep_aic_set <- function(aic_obj){
  aic_df <- as.data.frame(aic_obj)

  if ("Modnames" %in% names(aic_df)) {
    aic_df <- aic_df |> mutate(model_id = Modnames)
  } else {
    aic_df <- tibble::rownames_to_column(aic_df, "model_id")
  }

  if (!("AICc" %in% names(aic_df))) stop("Could not find an AICc column in the AIC table.")

  if (!("K" %in% names(aic_df))) {
    if ("k" %in% names(aic_df)) aic_df$K <- aic_df$k
    if ("df" %in% names(aic_df)) aic_df$K <- aic_df$df
  }

  if (!("K" %in% names(aic_df))) stop("Could not find K (number of parameters) in the AIC table.")

  aic_df |>
  mutate(
    AICc = as.numeric(AICc),

```

```

  `ΔAICc` = AICc - min(AICc, na.rm = TRUE),
  w = {
    rel <- exp(-0.5 * (AICc - min(AICc, na.rm = TRUE)))
    rel / sum(rel, na.rm = TRUE)
  }
) |>
select(model_id, K, AICc, `ΔAICc`, w) |>
arrange(`ΔAICc`)
}

get_deviance_original <- function(model_id, aic_obj, cand_list){
  aic_df <- as.data.frame(aic_obj)

  if ("Modnames" %in% names(aic_df)) {
    aic_df$model_id <- aic_df$Modnames
  } else if (!"model_id" %in% names(aic_df)) {
    aic_df <- tibble::rownames_to_column(aic_df, "model_id")
  }

  row <- aic_df[aic_df$model_id == model_id, , drop = FALSE]

  if (nrow(row) == 1) {
    if ("Deviance" %in% names(row) && is.finite(row$Deviance)) return(as.numeric(row$Deviance))
    if ("LL" %in% names(row) && is.finite(row$LL)) return(as.numeric(row$LL))
  }

  as.numeric(-2 * stats::logLik(cand_list[[model_id]]))
}

build_df <- function(tab, cand_list, aic_obj){
  tab |>
  mutate(
    Deviance = sapply(model_id, function(mn) get_deviance_original(mn, aic_obj, cand_list)),
    `R²` = sapply(model_id, function(mn) summary(cand_list[[mn]])$r.squared),
    `slope (SE)` = sapply(model_id, function(mn) get_slope_se_single_weather(cand_list[[mn]], mn)),
    Group = factor(model_group(model_id), levels = group_order),
    Model = label_model(model_id),
    K = as.integer(K)
  ) |>
  mutate(
    Deviance = round(Deviance, 2),
    AICc = round(AICc, 2),
    `ΔAICc` = round(`ΔAICc`, 2),
    w = round(w, 3),
    `R²` = round(`R²`, 3)
  ) |>
  arrange(Group, `ΔAICc`) |>
  select(Group, Model, K, Deviance, AICc, `ΔAICc`, w, `R²`, `slope (SE)`)
}

add_headers <- function(df, trend_label){
  df <- df |> mutate(Group = factor(Group, levels = group_order))

  groups_here <- levels(droplevels(df$Group))
  groups_here <- groups_here[groups_here %in% unique(as.character(df$Group))]

  empty_row <- df[0, ]

  trend_row <- empty_row
  trend_row[1, ] <- NA
  trend_row$Group <- NA
  trend_row$Model <- trend_label

  out <- trend_row
  is_trend <- TRUE
  is_group <- FALSE

```

```

for (g in groups_here) {
  gh <- empty_row
  gh[1, ] <- NA
  gh$Group <- NA
  gh$Model <- g

  rows_g <- df |> filter(as.character(Group) == g) |> mutate(Group = NA)

  out <- bind_rows(out, gh, rows_g)
  is_trend <- c(is_trend, FALSE, rep(FALSE, nrow(rows_g)))
  is_group <- c(is_group, TRUE, rep(FALSE, nrow(rows_g)))
}

df_print <- out |> select(-Group)
df_print[is.na(df_print)] <- ""

list(
  df_print = df_print,
  trend_rows = which(is_trend),
  group_rows = which(is_group)
)
}

tab_wy <- prep_aic_set(aic_effort_with_year)
tab_ny <- prep_aic_set(aic_effort_no_year)

df_wy <- build_df(tab_wy, cand_effort_with_year, aic_effort_with_year)
df_ny <- build_df(tab_ny, cand_effort_no_year, aic_effort_no_year)

hdr_wy <- add_headers(df_wy, "a) Models including year trend")
hdr_ny <- add_headers(df_ny, "b) Models without year trend")

df_print_1a <- hdr_wy$df_print
df_print_1b <- hdr_ny$df_print

make_ft <- function(df_print, trend_rows, group_rows){

  ft <- flextable(df_print) |>
    font(fontname = "Times New Roman", part = "all") |>
    fontsize(size = 12, part = "all") |>
    bold(part = "header") |>
    bg(part = "header", bg = "#f2f2f2") |>
    align(j = 1, align = "left", part = "all") |>
    align(j = 2:ncol(df_print), align = "center", part = "all") |>
    bg(i = trend_rows, bg = "#d9d9d9", part = "body") |>
    bold(i = trend_rows, part = "body") |>
    bg(i = group_rows, bg = "#e6e6e6", part = "body") |>
    bold(i = group_rows, part = "body") |>
    autofit()

  ft <- flextable::compose(
    ft,
    i = trend_rows,
    j = 1,
    part = "body",
    value = as_paragraph(
      as_chunk(
        df_print[trend_rows, 1],
        props = fp_text(underline = TRUE, bold = TRUE)
      )
    )
  )

  ft
}

```



```

ft_1a <- make_ft(df_print_1a, hdr_wy$trend_rows, hdr_wy$group_rows)
ft_1b <- make_ft(df_print_1b, hdr_ny$trend_rows, hdr_ny$group_rows)

caption_1a <- paste0(
  "Table 1a. Model selection results relating to annual weather variation and the number of active nests at Dronfield
  ↪ (1993–2024) for models including a linear year effect (corresponding to Fig. 1a). ",
  "Models were fitted using linear regressions. A '+' indicates additive effects and '×' indicates an interaction. ",
  "K is the number of parameters; AICc is Akaike's Information Criterion; ΔAICc is relative to the best model within
  ↪ this model set; w is the Akaike weight; R2 is variance explained. ",
  "Slopes (SE) are reported for simple predictor models only."
)

caption_1b <- paste0(
  "Table 1b. Model selection results relating to annual weather variation and the number of active nests at Dronfield
  ↪ (1993–2024) for models excluding a linear year effect (corresponding to Figs. 1b–c). ",
  "Models were fitted using linear regressions. A '+' indicates additive effects and '×' indicates an interaction. ",
  "K is the number of parameters; AICc is Akaike's Information Criterion; ΔAICc is relative to the best model within
  ↪ this model set; w is the Akaike weight; R2 is variance explained. ",
  "Slopes (SE) are reported for simple predictor models only."
)

doc <- read_docx() |>
  body_add_par(caption_1a, style = "Normal") |>
  body_add_par("", style = "Normal") |>
  body_add_flextable(ft_1a) |>
  body_add_par("", style = "Normal") |>
  body_add_par(caption_1b, style = "Normal") |>
  body_add_par("", style = "Normal") |>
  body_add_flextable(ft_1b)

print(doc, target = "Table_1a_1b_Active_Nests.docx")

```

B.3 Objective 2: Breeding success

B.3.1 Candidate model set

```

objective2_table <- tribble(
  ~`Hypothesis / justification`, ~Model,

  "**Temp**", "",
  "Extreme heat events cause chick heat stress and dehydration, reducing fledging success",
  "success ~ max_temp + (1 | year)",
  "Cold extremes increase thermoregulatory demand in chicks, increasing mortality risk",
  "success ~ min_temp + (1 | year)",

  "**Rain**", "",
  "Prolonged rainfall reduces adult foraging efficiency, limiting food delivery to chicks and reducing breeding
  ↪ success",
  "success ~ rain_days + (1 | year)",
  "Intense rainfall events cause acute nest disturbance and chick exposure, reducing breeding success",
  "success ~ max_rain + (1 | year)",
  "Overall wet years reduce provisioning efficiency across the breeding season, lowering breeding success",
  "success ~ total_rain + (1 | year)",

  "**Wind**", "",
  "Extreme winds increase nest exposure and disrupt provisioning, reducing chick survival",
  "success ~ max_wind + (1 | year)",

  "**Hail**", "",
  "Severe hail events increase nest destruction and chick mortality, reducing breeding success",
  "success ~ hail_days + (1 | year)",

  "**Joint effects**", "",

```

```

"Severe storm exposure increases nest destruction and chick mortality, reducing breeding success",
"success ~ hail_days + max_wind + (1 | year)",
"Prolonged wet conditions combined with storm events increase chick exposure and limit provisioning, reducing
  ↳ breeding success",
"success ~ rain_days + hail_days + (1 | year)",
"Heat stress combined with prolonged wet conditions reduces chick thermoregulation and food delivery, lowering
  ↳ breeding success",
"success ~ max_temp + rain_days + (1 | year)",

"**Interactive effects**", "",
"The negative effect of heat stress on breeding success is amplified during persistently wet conditions that limit
  ↳ adult foraging",
"success ~ max_temp * rain_days + (1 | year)",
"Cold stress on chicks is amplified during extreme wind events due to increased exposure and heat loss, reducing
  ↳ breeding success",
"success ~ min_temp * max_wind + (1 | year)"
)

knitr::kable(
  objective2_table,
  align = c("l", "l"),
  caption = "Objective 2 candidate model set for breeding success. All models are fitted as binomial GLMMs:
    ↳ cbind(success_nests, failed_nests) with a random intercept for year (1 | year)."
)

```

Table 3: Objective 2 candidate model set for breeding success. All models are fitted as binomial GLMMs: cbind(success_nests, failed_nests) with a random intercept for year (1 | year).

Hypothesis / justification	Model
Temp	
Extreme heat events cause chick heat stress and dehydration, reducing fledging success	success ~ max_temp + (1 year)
Cold extremes increase thermoregulatory demand in chicks, increasing mortality risk	success ~ min_temp + (1 year)
Rain	
Prolonged rainfall reduces adult foraging efficiency, limiting food delivery to chicks and reducing breeding success	success ~ rain_days + (1 year)
Intense rainfall events cause acute nest disturbance and chick exposure, reducing breeding success	success ~ max_rain + (1 year)
Overall wet years reduce provisioning efficiency across the breeding season, lowering breeding success	success ~ total_rain + (1 year)
Wind	
Extreme winds increase nest exposure and disrupt provisioning, reducing chick survival	success ~ max_wind + (1 year)
Hail	
Severe hail events increase nest destruction and chick mortality, reducing breeding success	success ~ hail_days + (1 year)
Joint effects	
Severe storm exposure increases nest destruction and chick mortality, reducing breeding success	success ~ hail_days + max_wind + (1 year)
Prolonged wet conditions combined with storm events increase chick exposure and limit provisioning, reducing breeding success	success ~ rain_days + hail_days + (1 year)
Heat stress combined with prolonged wet conditions reduces chick thermoregulation and food delivery, lowering breeding success	success ~ max_temp + rain_days + (1 year)
Interactive effects	
The negative effect of heat stress on breeding success is amplified during persistently wet conditions that limit adult foraging	success ~ max_temp * rain_days + (1 year)
Cold stress on chicks is amplified during extreme wind events due to increased exposure and heat loss, reducing breeding success	success ~ min_temp * max_wind + (1 year)

B.3.2 Data used

```
success_df <- vultures.clean |>
  dplyr::mutate(year = lubridate::year(ringing.date)) |>
  dplyr::filter(!is.na(year)) |>
  dplyr::group_by(year) |>
  dplyr::summarise(
    active_nests = dplyr::n(),
    success_nests = sum(!is.na(laying.date)),
    failed_nests = active_nests - success_nests,
    success_prop = success_nests / active_nests,
    .groups = "drop"
  ) |>
  dplyr::filter(active_nests > 0) |>
  dplyr::left_join(weather_yearly, by = c("year" = "YEAR")) |>
  dplyr::mutate(
    year = factor(year),

    z_max_temp = as.numeric(scale(max_temp)),
    z_min_temp = as.numeric(scale(min_temp)),
    z_rain_days = as.numeric(scale(rain_days)),
    z_max_rain = as.numeric(scale(max_rain)),
    z_total_rain = as.numeric(scale(total_rain)),
    z_max_wind = as.numeric(scale(max_wind)),
    z_hail_days = as.numeric(scale(hail_days))
  )

success_cc <- success_df |>
  dplyr::select(
    year, success_nests, failed_nests,
    z_max_temp, z_min_temp,
    z_rain_days, z_max_rain, z_total_rain,
    z_max_wind, z_hail_days
  ) |>
  tidyr::drop_na()

nrow(success_df)
```

[1] 32

```
nrow(success_cc)
```

[1] 32

B.3.3 Checking overdispersion

```
success_year_glm <- glm(
  cbind(success_nests, failed_nests) ~ year,
  family = binomial,
  data = success_cc
)

summary(success_year_glm)
```

Call:

```
glm(formula = cbind(success_nests, failed_nests) ~ year, family = binomial,
```

```
data = success_cc)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.8109	0.3469	2.337	0.019421	*
year1994	-0.2179	0.4408	-0.494	0.621128	
year1995	-0.4745	0.4844	-0.979	0.327358	
year1996	-0.5804	0.4322	-1.343	0.179326	
year1997	0.3567	0.4626	0.771	0.440718	
year1998	-0.1178	0.4342	-0.271	0.786200	
year1999	-0.6286	0.4260	-1.476	0.140055	
year2000	-0.2131	0.4368	-0.488	0.625683	
year2001	0.3060	0.4406	0.695	0.487296	
year2002	-1.2048	0.4220	-2.855	0.004306	**
year2003	-0.5232	0.4303	-1.216	0.224016	
year2004	-0.7484	0.4277	-1.750	0.080148	.
year2005	-0.1398	0.4314	-0.324	0.745935	
year2006	-0.4925	0.4175	-1.179	0.238221	
year2007	-0.7655	0.4072	-1.880	0.060159	.
year2008	-0.6033	0.4085	-1.477	0.139685	
year2009	0.1301	0.4553	0.286	0.775132	
year2010	-1.1727	0.4048	-2.897	0.003768	**
year2011	-0.3756	0.4127	-0.910	0.362696	
year2012	-1.5841	0.4258	-3.721	0.000199	***
year2013	-0.9612	0.4131	-2.327	0.019975	*
year2014	-0.1919	0.4054	-0.473	0.635949	
year2015	-0.5401	0.3994	-1.352	0.176351	
year2016	-0.8311	0.4010	-2.073	0.038191	*
year2017	-0.6252	0.3971	-1.574	0.115429	
year2018	-0.7224	0.3948	-1.830	0.067265	.
year2019	-0.4745	0.3981	-1.192	0.233311	
year2020	-1.0986	0.4025	-2.729	0.006349	**
year2021	-1.0918	0.4096	-2.665	0.007691	**
year2022	-1.0838	0.3936	-2.754	0.005892	**
year2023	-0.6796	0.3915	-1.736	0.082613	.
year2024	-0.8689	0.3865	-2.248	0.024564	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1.0769e+02 on 31 degrees of freedom
Residual deviance: -5.0404e-14 on 0 degrees of freedom
AIC: 216.3

Number of Fisher Scoring iterations: 3

```
performance::check_overdispersion(success_year_glm)
```

Overdispersion test

dispersion ratio = 0.877
p-value = 0.304

B.3.4 Fitting the models

```
library(lme4)

success_df <- success_df |> dplyr::mutate(year = factor(year))

# Baseline (null) model

m0_success <- glmer(
  cbind(success_nests, failed_nests) ~ 1 + (1 | year),
  family = binomial,
  data = success_df
)

# Single predictor models

m_succ_maxT <- glmer(
  cbind(success_nests, failed_nests) ~ z_max_temp + (1 | year),
  family = binomial, data = success_df
)

m_succ_minT <- glmer(
  cbind(success_nests, failed_nests) ~ z_min_temp + (1 | year),
  family = binomial, data = success_df
)

m_succ_rainD <- glmer(
  cbind(success_nests, failed_nests) ~ z_rain_days + (1 | year),
  family = binomial, data = success_df
)

m_succ_maxR <- glmer(
  cbind(success_nests, failed_nests) ~ z_max_rain + (1 | year),
  family = binomial, data = success_df
)

m_succ_totR <- glmer(
  cbind(success_nests, failed_nests) ~ z_total_rain + (1 | year),
  family = binomial, data = success_df
)

m_succ_maxW <- glmer(
  cbind(success_nests, failed_nests) ~ z_max_wind + (1 | year),
  family = binomial, data = success_df
)

m_succ_hail <- glmer(
  cbind(success_nests, failed_nests) ~ z_hail_days + (1 | year),
  family = binomial, data = success_df
)

# Joint effects

m_succ_hailW <- glmer(
  cbind(success_nests, failed_nests) ~ z_hail_days + z_max_wind + (1 | year),
  family = binomial, data = success_df
)

m_succ_rainH <- glmer(
  cbind(success_nests, failed_nests) ~ z_rain_days + z_hail_days + (1 | year),
  family = binomial, data = success_df
)

m_succ_heatR <- glmer(
  cbind(success_nests, failed_nests) ~ z_max_temp + z_rain_days + (1 | year),
  family = binomial, data = success_df
)
```

```

)

# Interactions

m_succ_int_heat_rain <- glmer(
  cbind(success_nests, failed_nests) ~ z_max_temp * z_rain_days + (1 | year),
  family = binomial, data = success_df
)

m_succ_int_cold_wind <- glmer(
  cbind(success_nests, failed_nests) ~ z_min_temp * z_max_wind + (1 | year),
  family = binomial, data = success_df
)

```

B.3.5 AIC model selection

```

cand_success <- list(
  null_re      = m0_success,

  max_temp     = m_succ_maxT,
  min_temp     = m_succ_minT,
  rain_days    = m_succ_rainD,
  max_rain     = m_succ_maxR,
  total_rain   = m_succ_totR,
  max_wind     = m_succ_maxW,
  hail_days    = m_succ_hail,

  hail_maxW    = m_succ_hailW,
  rain_hail    = m_succ_rainH,
  maxT_rainD   = m_succ_heatR,

  int_heatRain = m_succ_int_heat_rain,
  int_coldWind = m_succ_int_cold_wind
)

aic_success <- aictab(
  cand.set = cand_success,
  modnames = names(cand_success)
)

aic_success

```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
min_temp	3	226.93	0.00	0.43	0.43	-110.03
null_re	2	229.45	2.52	0.12	0.56	-112.52
hail_days	3	230.45	3.53	0.07	0.63	-111.80
max_temp	3	230.85	3.92	0.06	0.69	-111.99
max_rain	3	231.01	4.09	0.06	0.75	-112.08
rain_days	3	231.61	4.68	0.04	0.79	-112.38
total_rain	3	231.65	4.72	0.04	0.83	-112.40
max_wind	3	231.73	4.81	0.04	0.87	-112.44
int_coldWind	5	232.04	5.11	0.03	0.91	-109.87
maxT_rainD	4	232.10	5.18	0.03	0.94	-111.31
rain_hail	4	232.30	5.38	0.03	0.97	-111.41
hail_maxW	4	232.67	5.74	0.02	0.99	-111.59
int_heatRain	5	234.93	8.01	0.01	1.00	-111.31

```
vc_year <- function(mod) {
  vc <- as.data.frame(VarCorr(mod))
  vc$vcov[vc$grp == "year"][1]
}

v0 <- vc_year(m0_success)
v_minT <- vc_year(m_succ_minT)

prop_between_year_explained <- (v0 - v_minT) / v0
prop_between_year_explained
```

```
[1] 0.2098479
```

B.3.6 Best models

```
aic_df_success <- as.data.frame(aic_success)

best_success_id <- if ("Modnames" %in% names(aic_df_success)) {
  aic_df_success$Modnames[1]
} else {
  rownames(aic_df_success)[1]
}

best_success_mod <- cand_success[[best_success_id]]

best_success_id
```

```
[1] "min_temp"
```

```
summary(best_success_mod)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: cbind(success_nests, failed_nests) ~ z_min_temp + (1 | year)
Data: success_df
```

AIC	BIC	logLik	-2*log(L)	df.resid
226.1	230.5	-110.0	220.1	29

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.2181	-0.3386	-0.1177	0.5078	1.3408

Random effects:

Groups	Name	Variance	Std.Dev.
year	(Intercept)	0.1021	0.3195

Number of obs: 32, groups: year, 32

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.21436	0.07017	3.055	0.00225 **
z_min_temp	0.16711	0.07222	2.314	0.02066 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
(Intr)
z_min_temp -0.005

B.3.7 Adding population size to the best supported models:

```
success_df <- success_df |>  
  mutate(z_active_nests = scale(active_nests))
```

```
m_succ_pop <- glmer(  
  cbind(success_nests, failed_nests) ~ z_active_nests + (1 | year),  
  family = binomial,  
  data = success_df  
)
```

```
m_succ_minT_pop <- glmer(  
  cbind(success_nests, failed_nests) ~ z_min_temp + z_active_nests + (1 | year),  
  family = binomial,  
  data = success_df  
)
```

```
summary(m_succ_pop)
```

Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial (logit)
Formula: cbind(success_nests, failed_nests) ~ z_active_nests + (1 | year)
Data: success_df

AIC	BIC	logLik	-2*log(L)	df.resid
224.6	229.0	-109.3	218.6	29

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.5918	-0.3451	0.0927	0.3398	1.2347

Random effects:

Groups	Name	Variance	Std.Dev.
year	(Intercept)	0.09762	0.3124

Number of obs: 32, groups: year, 32

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.23572	0.06936	3.399	0.000677 ***
z_active_nests	-0.18646	0.07062	-2.640	0.008286 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
(Intr)
z_actv_nsts -0.117


```
summary(m_succ_minT)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: cbind(success_nests, failed_nests) ~ z_min_temp + (1 | year)
Data: success_df
```

AIC	BIC	logLik	-2*log(L)	df.resid
226.1	230.5	-110.0	220.1	29

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.2181	-0.3386	-0.1177	0.5078	1.3408

Random effects:

Groups	Name	Variance	Std.Dev.
year	(Intercept)	0.1021	0.3195

Number of obs: 32, groups: year, 32

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.21436	0.07017	3.055	0.00225 **
z_min_temp	0.16711	0.07222	2.314	0.02066 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)
z_min_temp -0.005

```
summary(m_succ_minT_pop)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: cbind(success_nests, failed_nests) ~ z_min_temp + z_active_nests +
(1 | year)
Data: success_df
```

AIC	BIC	logLik	-2*log(L)	df.resid
218.5	224.4	-105.3	210.5	28

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.67994	-0.44118	0.07867	0.37818	1.53542

Random effects:

Groups	Name	Variance	Std.Dev.
year	(Intercept)	0.0643	0.2536

Number of obs: 32, groups: year, 32

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.23483	0.06137	3.826	0.00013	***
z_min_temp	0.19152	0.06372	3.006	0.00265	**
z_active_nests	-0.20533	0.06279	-3.270	0.00108	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	z_mn_t
z_min_temp	0.012	
z_actv_nsts	-0.147	-0.125

```
aictab(list(
  min_temp      = m_succ_minT,
  min_temp_pop  = m_succ_minT_pop,
  pop_only      = m_succ_pop
))
```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
min_temp_pop	4	220.02	0.00	0.91	0.91	-105.27
pop_only	3	225.45	5.44	0.06	0.97	-109.30
min_temp	3	226.93	6.91	0.03	1.00	-110.03

```
summary(m_succ_minT_pop)
```

Generalized linear mixed model fit by maximum likelihood (Laplace Approximation) [glmerMod]

Family: binomial (logit)

Formula: cbind(success_nests, failed_nests) ~ z_min_temp + z_active_nests + (1 | year)

Data: success_df

AIC	BIC	logLik	-2*log(L)	df.resid
218.5	224.4	-105.3	210.5	28

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.67994	-0.44118	0.07867	0.37818	1.53542

Random effects:

Groups	Name	Variance	Std.Dev.
year	(Intercept)	0.0643	0.2536

Number of obs: 32, groups: year, 32

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.23483	0.06137	3.826	0.00013	***
z_min_temp	0.19152	0.06372	3.006	0.00265	**
z_active_nests	-0.20533	0.06279	-3.270	0.00108	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr) z_mn_t

z_min_temp 0.012

z_actv_nsts -0.147 -0.125

```
vc_year <- function(mod) {  
  vc <- as.data.frame(VarCorr(mod))  
  vc$vcov[vc$grp == "year"][1]  
}  
  
v0 <- vc_year(m0_success)  
v1 <- vc_year(m_succ_minT_pop)  
  
prop_between_year_explained <- (v0 - v1) / v0  
prop_between_year_explained
```

[1] 0.5023207

```
v_minT <- vc_year(m_succ_minT)  
  
prop_minT <- (v0 - v_minT) / v0  
prop_minT
```

[1] 0.2098479

```
coef(summary(m_succ_minT_pop))["z_min_temp", c("Estimate", "Std. Error")]
```

	Estimate	Std. Error
	0.19151634	0.06371546

```
v_pop <- vc_year(m_succ_pop)  
  
prop_pop <- (v0 - v_pop) / v0  
prop_pop
```

[1] 0.2443964

```
vc_year <- function(mod) {  
  vc <- as.data.frame(VarCorr(mod))  
  out <- vc$vcov[vc$grp == "year"][1]  
  if (length(out) == 0) NA_real_ else out  
}  
  
slope_se_single <- function(mod, digits = 3){  
  fe <- lme4::fixef(mod)  
  
  if (length(fe) == 2) {  
    cf <- summary(mod)$coefficients  
    paste0(  
      round(cf[2, 1], digits),  
      " (", round(cf[2, 2], digits), ")"  
    )  
  } else {
```

```

    ""
  }
}

annual_var_expl_factory <- function(null_mod){
  v0 <- vc_year(null_mod)
  function(mod){
    v_mod <- vc_year(mod)
    if (is.finite(v_mod) && is.finite(v0) && v0 > 0) max(0, (v0 - v_mod)/v0) else NA_real_
  }
}

aic_df <- as.data.frame(aic_success)

if ("Modnames" %in% names(aic_df)) {
  aic_df <- aic_df |> mutate(model_id = Modnames)
} else {
  aic_df <- tibble::rownames_to_column(aic_df, "model_id")
}

table_success <- aic_df |>
  rename(`ΔAICc` = Delta_AICc, w = AICcWt) |>
  select(model_id, K, AICc, `ΔAICc`, w) |>
  arrange(`ΔAICc`)

var_expl_fun <- annual_var_expl_factory(m0_success)

stats_df <- table_success |>
  mutate(
    `Between-year variance explained` =
      sapply(model_id, function(mn) var_expl_fun(cand_success[[mn]])),
    `slope (SE)` =
      sapply(model_id, function(mn) slope_se_single(cand_success[[mn]]))
  )

table_success2 <- table_success |>
  left_join(stats_df |> select(model_id, `Between-year variance explained`, `slope (SE)`),
    by = "model_id") |>
  mutate(
    AICc = round(AICc, 2),
    `ΔAICc` = round(`ΔAICc`, 2),
    w = round(w, 3),
    `Between-year variance explained` = round(`Between-year variance explained`, 3)
  )

group_order <- c("Baseline", "Temperature", "Rain", "Wind", "Hail", "Joint effects", "Interactions")

model_group_success <- function(m){
  dplyr::case_when(
    m %in% c("null_re") ~ "Baseline",
    m %in% c("max_temp", "min_temp") ~ "Temperature",
    m %in% c("rain_days", "max_rain", "total_rain") ~ "Rain",
    m %in% c("max_wind") ~ "Wind",
    m %in% c("hail_days") ~ "Hail",
    m %in% c("hail_maxW", "rain_hail", "maxT_rainD") ~ "Joint effects",
    m %in% c("int_heatRain", "int_coldWind") ~ "Interactions",
    TRUE ~ "Joint effects"
  )
}

pretty_model_success <- function(x){
  dplyr::recode(
    x,
    "null_re" = "Intercept only",
    "max_temp" = "Maximum temperature",
    "min_temp" = "Minimum temperature",
    "rain_days" = "Rainy days",

```

```

    "max_rain"      = "Maximum daily rainfall",
    "total_rain"    = "Total rainfall",
    "max_wind"      = "Maximum wind speed",
    "hail_days"     = "Hail days",
    "hail_maxW"     = "Hail days + max wind speed",
    "rain_hail"     = "Rainy days + hail days",
    "maxT_rainD"    = "Max temperature + rainy days",
    "int_heatRain"  = "Max temperature × rainy days",
    "int_coldWind"  = "Min temperature × max wind speed",
    .default = x
  )
}

df <- table_success2 |>
  mutate(
    Group = factor(model_group_success(model_id), levels = group_order),
    Model = pretty_model_success(model_id)
  ) |>
  arrange(Group, `ΔAICc`) |>
  select(Group, Model, K, AICc, `ΔAICc`, w, `Between-year variance explained`, `slope (SE)`)

groups <- levels(droplevels(df$Group))
groups <- groups[groups %in% unique(as.character(df$Group))]

make_group_block <- function(g) {
  header <- df[0, ]
  header[1, ] <- NA
  header$Group <- g
  header$Model <- g
  header$is_group <- TRUE

  rows <- df |> filter(as.character(Group) == g)
  rows$is_group <- FALSE

  bind_rows(header, rows)
}

df2 <- bind_rows(lapply(groups, make_group_block))
group_rows <- which(df2$is_group)

df_print <- df2 |> select(-Group, -is_group)

ft <- flextable(df_print) |>
  font(fontname = "Times New Roman", part = "all") |>
  fontsize(size = 12, part = "all") |>
  bold(part = "header") |>
  bg(part = "header", bg = "#f2f2f2") |>
  align(j = 1, align = "left", part = "all") |>
  align(j = 2:ncol(df_print), align = "center", part = "all") |>
  bg(i = group_rows, bg = "#e6e6e6", part = "body") |>
  bold(i = group_rows, part = "body") |>
  autofit()

doc <- read_docx() |> body_add_flextable(ft)
print(doc, target = "Table_2_Breeding_Success_REBUILT.docx")

```

```

sens_tbl <- tibble(
  Model = c(
    "Minimum temperature",
    "Population size",
    "Minimum temperature + population size"
  ),
  `ΔAICc` = c(6.91, 5.44, 0.00),
  `AIC weight` = c(0.03, 0.06, 0.91),
  `Between-year variance explained` = c(0.21, 0.24, 0.50)
)

```

```

ft <- flextable(sens_tbl)

ft <- ft |>
  font(fontname = "Times New Roman", part = "all") |>
  fontsize(size = 12, part = "all") |>
  bold(part = "header") |>
  bg(part = "header", bg = "#f2f2f2") |>
  align(j = 1, align = "left", part = "all") |>
  align(j = 2:ncol(sens_tbl), align = "center", part = "all") |>
  autofit()

doc <- read_docx() |>
  body_add_flextable(ft)

print(doc, target = "Sensitivity_Table.docx")

```

B.3.8 Combines table

```

sens_tbl <- tibble(
  Model = c(
    "Minimum temperature",
    "Population size",
    "Minimum temperature + population size"
  ),
  `ΔAICc` = c(6.91, 5.44, 0.00),
  w = c(0.03, 0.06, 0.91),
  `Between-year variance explained` = c(0.21, 0.24, 0.50)
)

align_to_template <- function(x, template_names) {
  missing <- setdiff(template_names, names(x))
  if (length(missing) > 0) x[missing] <- NA
  x <- x[, template_names, drop = FALSE]
  x
}

variance_targets <- c(
  "Between-year variance explained",
  "Annual variance explained",
  "Variance explained",
  "Deviance explained"
)

target_var_col <- intersect(variance_targets, names(df_print))[1]

sens_tbl2 <- sens_tbl

if (!is.na(target_var_col) && target_var_col != "Between-year variance explained") {
  sens_tbl2 <- sens_tbl2 |>
    rename(!target_var_col := `Between-year variance explained`) |>
    select(-`Between-year variance explained`)
}

sens_tbl2 <- align_to_template(sens_tbl2, names(df_print))

sens_header <- df_print[0, ]
sens_header[1, ] <- NA
sens_header$Model <- "A posteriori models: population size"
sens_header$is_group <- TRUE

main_tbl <- df_print |> mutate(is_group = FALSE)
sens_tbl2 <- sens_tbl2 |> mutate(is_group = FALSE)

```

```

df_final_print <- bind_rows(
  main_tbl,
  sens_header,
  sens_tbl2
) |>
  select(-is_group)

group_rows_combined <- c(group_rows, nrow(main_tbl) + 1)

ft <- flextable(df_final_print) |>
  font(fontname = "Times New Roman", part = "all") |>
  fontsize(size = 12, part = "all") |>
  bold(part = "header") |>
  bg(part = "header", bg = "#f2f2f2") |>
  align(j = 1, align = "left", part = "all") |>
  align(j = 2:ncol(df_final_print), align = "center", part = "all") |>
  bg(i = group_rows_combined, bg = "#e6e6e6", part = "body") |>
  bold(i = group_rows_combined, part = "body") |>
  autofit()

doc <- read_docx() |> body_add_flextable(ft)
print(doc, target = "Table_2_Breeding_Success_WITH_PostHoc.docx")

```

B.4 Objective 3: Breeding timing of successful nests

B.4.1 DAGS

B.4.2 Scenario A

```

dir.create("figures", showWarnings = FALSE)

export_dag_pdf <- function(dag, filename, w = 8, h = 4) {
  svg_txt <- DiagrammeRsvg::export_svg(dag)
  rsvg::rsvg_pdf(
    charToRaw(svg_txt),
    file = file.path("figures", filename),
    width = w, height = h
  )
  file.info(file.path("figures", filename))[, "size"]
}

timing_simple_dag <- DiagrammeR::grViz("
digraph timing_simple {
  graph [
    layout = neato,
    splines = false,
    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]
}

```

```

]

W      [label = 'W',  shape = box,    pos = '1.4,1.5!']

T [label = 'T',
    shape = circle,
    fixedsize = true,
    width = 1,
    pos = '2.8,1.5!']
]

S      [label = 'S',  shape = box,    pos = '1.4,0!']
Tstar [label = 'T*', shape = box,    pos = '2.8,0!']

W -> T
T -> Tstar
S -> Tstar
}
")

export_dag_pdf(timing_simple_dag, "timing_simple_dag.pdf", 8, 4)

```

[1] 6396

B.4.3 Scenario B

```

timing_B_dag <- DiagrammeR::grViz("
digraph timing_B {
  graph [
    layout = neato,
    splines = false,
    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]

  W      [label = 'W',  shape = box,    pos = '1.0,1.5!']

  T [label = 'T',
    shape = circle,
    fixedsize = true,
    width = 1,
    pos = '2.8,1.5!']
]

S      [label = 'S',  shape = box,    pos = '1.0,0!']
Tstar [label = 'T*', shape = box,    pos = '2.8,0!']

W -> T
W -> S
T -> Tstar
S -> Tstar
}

```



```
)

export_dag_pdf(timing_B_dag, "timing_B_dag.pdf", 8, 4)
```

```
[1] 6445
```

B.4.4 Scenario C

```
timing_C_dag <- DiagrammeR::grViz("
digraph timing_C {
  graph [
    layout = neato,
    splines = false,
    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]

  W      [label = 'W',  shape = box,    pos = '1.0,1.5!']

  T [label = 'T',
    shape = circle,
    fixedsize = true,
    width = 1,
    pos = '2.8,1.5!']
  ]

  S      [label = 'S',  shape = box,    pos = '1.0,0!']
  Tstar [label = 'T*', shape = box,    pos = '2.8,0!']

  W -> T
  T -> Tstar
  S -> Tstar
  T -> S
}
")

export_dag_pdf(timing_C_dag, "timing_C_dag.pdf", 8, 4)
```

```
[1] 6456
```

B.4.5 Scenario D

```
timing_D_dag <- DiagrammeR::grViz("
digraph timing_D {
  graph [
    layout = neato,
    splines = false,
```

```

    overlap = false
  ]

  node [
    fontname = Helvetica,
    fontsize = 22,
    penwidth = 2,
    margin = 0.25
  ]

  edge [
    penwidth = 1.8,
    arrowsize = 0.55
  ]

  W      [label = 'W',  shape = box,    pos = '1.0,1.5!']

  T [label = 'T',
    shape = circle,
    fixedsize = true,
    width = 1,
    pos = '2.8,1.5!']
  ]

  S      [label = 'S',  shape = box,    pos = '1.0,0!']
  Tstar [label = 'T*', shape = box,    pos = '2.8,0!']

  X [label = 'X',
    shape = circle,
    fixedsize = true,
    width = 0.9,
    pos = '2.0,0.75!']
  ]

  W -> T
  T -> Tstar
  S -> Tstar
  X -> T
  X -> S
}
")

export_dag_pdf(timing_D_dag, "timing_D_dag.pdf", 8, 4)

```

[1] 6804

B.4.6 Models

B.4.7 Candidate model set

```

library(tibble)
library(knitr)

objective3_table <- tribble(
  ~`Hypothesis / justification`, ~Model,

  # ---- Baseline ----
  "**Baseline**", "",
  "Baseline (null / intercept-only) model for breeding timing (laying date, day of year).",
  "lay_DOY ~ 1",

  # ---- Temperature (March cues) ----

```

```

**Temperature**", "",
"Warmer conditions in March may act as a cue for the initiation of breeding, advancing laying dates.",
"lay_DOY ~ meanT",

"Colder March conditions may delay breeding due to increased energetic costs and reduced adult condition.",
"lay_DOY ~ minT",

# ---- Rain (March cues) ----
**Rain**", "",
"Prolonged wet conditions in March may delay breeding by limiting foraging efficiency.",
"lay_DOY ~ rainD",

"Higher March rainfall may signal improved food availability and advance breeding timing.",
"lay_DOY ~ totR",

# ---- Wind (March cues) ----
**Wind**", "",
"Persistent windy conditions in March increase flight costs and may delay breeding timing.",
"lay_DOY ~ wind",

# ---- Lagged effects (previous year) ----
**Lagged effects**", "",
"Thermal conditions in the previous year may influence physiological state entering the breeding season and affect
↪ laying timing.",
"lay_DOY ~ lagMeanT",

"Higher rainfall in the previous year may improve food availability and adult condition entering the breeding
↪ season, allowing earlier laying.",
"lay_DOY ~ lagTotR",

"Prolonged wet conditions in the previous year may influence adult condition and carry over to affect laying
↪ timing.",
"lay_DOY ~ lagRainD",

# ---- Joint effects ----
**Joint effects**", "",
"March cues and previous-year rainfall may jointly influence breeding timing.",
"lay_DOY ~ cue_plus_lagR",

"Previous-year rainfall and March rainy days may jointly influence breeding timing.",
"lay_DOY ~ lagR_plus_cueRain",

# ---- Interactions ----
**Interactions**", "",
"The effect of March temperature cues on laying timing may depend on March rainfall conditions.",
"lay_DOY ~ int_TR",

"Carry-over effects from the previous year may modify responses to March temperature cues.",
"lay_DOY ~ lagR_x_meanT"
)

kable(
  objective3_table,
  align = c("l", "l"),
  caption = "Objective 3 candidate model set for breeding timing (laying date, day of year). March weather cues and
↪ lagged (previous-year) effects."
)

```

Table 4: Objective 3 candidate model set for breeding timing (laying date, day of year). March weather cues and lagged (previous-year) effects.

Hypothesis / justification	Model
Baseline	
Baseline (null / intercept-only) model for breeding timing (laying date, day of year).	lay_DOY ~ 1

Hypothesis / justification	Model
Temperature	
Warmer conditions in March may act as a cue for the initiation of breeding, advancing laying dates.	lay_DOY ~ meanT
Colder March conditions may delay breeding due to increased energetic costs and reduced adult condition.	lay_DOY ~ minT
Rain	
Prolonged wet conditions in March may delay breeding by limiting foraging efficiency.	lay_DOY ~ rainD
Higher March rainfall may signal improved food availability and advance breeding timing.	lay_DOY ~ totR
Wind	
Persistent windy conditions in March increase flight costs and may delay breeding timing.	lay_DOY ~ wind
Lagged effects	
Thermal conditions in the previous year may influence physiological state entering the breeding season and affect laying timing.	lay_DOY ~ lagMeanT
Higher rainfall in the previous year may improve food availability and adult condition entering the breeding season, allowing earlier laying.	lay_DOY ~ lagTotR
Prolonged wet conditions in the previous year may influence adult condition and carry over to affect laying timing.	lay_DOY ~ lagRainD
Joint effects	
March cues and previous-year rainfall may jointly influence breeding timing.	lay_DOY ~ cue_plus_lagR
Previous-year rainfall and March rainy days may jointly influence breeding timing.	lay_DOY ~ lagR_plus_cueRain
Interactions	
The effect of March temperature cues on laying timing may depend on March rainfall conditions.	lay_DOY ~ int_TR
Carry-over effects from the previous year may modify responses to March temperature cues.	lay_DOY ~ lagR_x_meanT

B.4.8 Timing data used

```

timing_df <- vultures.clean |>
  dplyr::filter(!is.na(laying.date)) |>
  dplyr::mutate(
    year      = as.integer(lubridate::year(laying.date)),
    lay_DOY   = lubridate::yday(laying.date)
  ) |>
  dplyr::filter(!is.na(year), year != 1900)

# One value per year
timing_by_year <- timing_df |>
  dplyr::group_by(year) |>
  dplyr::summarise(
    lay_DOY_median = median(lay_DOY, na.rm = TRUE),
    lay_DOY_mean   = mean(lay_DOY, na.rm = TRUE),
    n_nests        = sum(!is.na(lay_DOY)),
    first_DOY      = min(lay_DOY, na.rm = TRUE),
    last_DOY       = max(lay_DOY, na.rm = TRUE),
    .groups = "drop"
  ) |>
  dplyr::arrange(year)

timing_by_year

# A tibble: 32 x 6
  year lay_DOY_median lay_DOY_mean n_nests first_DOY last_DOY
<int>         <dbl>         <dbl>   <int>     <dbl>     <dbl>

```

1	1993	148.	148.	28	103	203
2	1994	159	162.	39	128	212
3	1995	156.	154.	22	132	171
4	1996	158	158.	35	129	209
5	1997	151	152.	46	132	190
6	1998	148	151.	45	122	188
7	1999	152	153.	37	136	192
8	2000	153	155.	41	124	201
9	2001	156.	158.	56	129	208
10	2002	154.	155.	30	20	185

i 22 more rows

```
ggplot(timing_by_year, aes(x = year, y = lay_DOY_median)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(x = "Year", y = "Median laying date (DOY)") +
  theme_bw() +
  theme(panel.grid = element_blank())
```

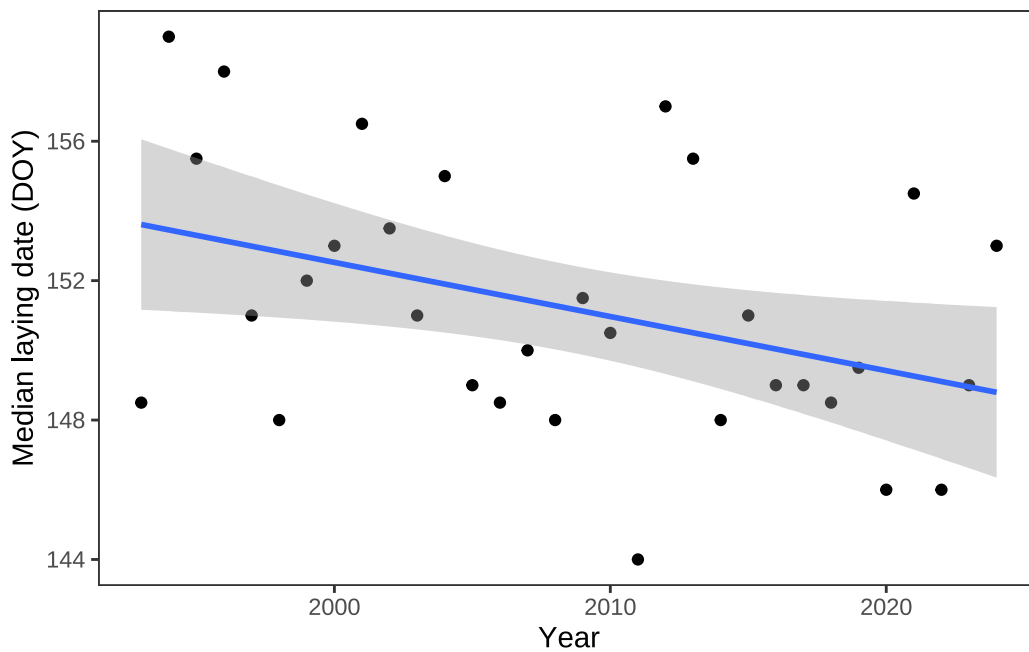


Figure 5: Variation in breeding timing (laying date) across the study period in Dronfield (1993-2024).

```
obj3_year_df <- timing_by_year |>
  left_join(weather_march_by_year, by = "year") |>
  left_join(
    weather_annual_by_year |>
      dplyr::select(year,
                    lag_mean_temp_annual,
                    lag_total_rain_annual,
                    lag_rain_days_annual),
    by = "year"
  ) |>
  arrange(year)

obj3_year_df
```

A tibble: 32 x 19

```

  year lay_DOY_median lay_DOY_mean n_nests first_DOY last_DOY mean_temp_win
<dbl>      <dbl>      <dbl>      <int>      <dbl>      <dbl>      <dbl>
1  1993          148.        148.        28        103        203        21.0
2  1994          159        162.        39        128        212        21.7
3  1995          156.        154.        22        132        171        22.1
4  1996          158        158.        35        129        209        22.3
5  1997          151        152.        46        132        190        19.8
6  1998          148        151.        45        122        188        22.2
7  1999          152        153.        37        136        192        25.4
8  2000          153        155.        41        124        201        22.3
9  2001          156.        158.        56        129        208        23.2
10 2002          154.        155.        30         20        185        23.5
# i 22 more rows
# i 12 more variables: max_temp_win <dbl>, min_temp_win <dbl>,
#   total_rain_win <dbl>, max_rain_win <dbl>, rain_days_win <int>,
#   mean_wind_win <dbl>, max_wind_win <dbl>, hail_days_win <dbl>, n_days <int>,
#   lag_mean_temp_annual <dbl>, lag_total_rain_annual <dbl>,
#   lag_rain_days_annual <int>

```

```

timing_by_year <- timing_df |>
  dplyr::filter(!is.na(year), year != 1900) |>
  dplyr::mutate(year = as.integer(year)) |>
  dplyr::group_by(year) |>
  dplyr::summarise(
    lay_DOY_median = median(lay_DOY, na.rm = TRUE),
    lay_DOY_mean   = mean(lay_DOY, na.rm = TRUE),
    n_nests        = sum(!is.na(lay_DOY)),
    first_DOY      = min(lay_DOY, na.rm = TRUE),
    last_DOY       = max(lay_DOY, na.rm = TRUE),
    .groups = "drop"
  ) |>
  dplyr::arrange(year)

```

```

vars_obj3 <- c(
  "year",
  "lay_DOY_median",
  "mean_temp_win", "min_temp_win",
  "rain_days_win", "total_rain_win",
  "mean_wind_win",
  "lag_mean_temp_annual",
  "lag_total_rain_annual",
  "lag_rain_days_annual"
)

obj3_cc <- obj3_year_df |>
  dplyr::select(all_of(vars_obj3)) |>
  tidyr::drop_na()

```

```
nrow(obj3_year_df)
```

```
[1] 32
```

```
summary(obj3_year_df$lay_DOY_median)
```

```

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
144.0  148.5   150.8   151.2  153.8   159.0

```

```
summary(obj3_year_df$mean_temp_win)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
18.70	21.40	22.03	22.10	22.94	25.41

```
cand_obj3 <- list(
  null = lm(lay_DOY_median ~ 1, data = obj3_cc),

  meanT = lm(lay_DOY_median ~ mean_temp_win, data = obj3_cc),
  minT = lm(lay_DOY_median ~ min_temp_win, data = obj3_cc),

  rainD = lm(lay_DOY_median ~ rain_days_win, data = obj3_cc),
  totR = lm(lay_DOY_median ~ total_rain_win, data = obj3_cc),

  wind = lm(lay_DOY_median ~ mean_wind_win, data = obj3_cc),

  lagMeanT = lm(lay_DOY_median ~ lag_mean_temp_annual, data = obj3_cc),
  lagTotR = lm(lay_DOY_median ~ lag_total_rain_annual, data = obj3_cc),
  lagRainD = lm(lay_DOY_median ~ lag_rain_days_annual, data = obj3_cc),

  cue_plus_lagR =
    lm(lay_DOY_median ~ mean_temp_win + rain_days_win + lag_total_rain_annual,
      data = obj3_cc),

  lagR_plus_cueRain =
    lm(lay_DOY_median ~ lag_total_rain_annual + rain_days_win,
      data = obj3_cc),

  int_TR =
    lm(lay_DOY_median ~ mean_temp_win * rain_days_win,
      data = obj3_cc),

  lagR_x_meanT =
    lm(lay_DOY_median ~ lag_total_rain_annual * mean_temp_win,
      data = obj3_cc)
)

aictab(cand.set = cand_obj3, modnames = names(cand_obj3))
```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
lagTotR	3	173.52	0.00	0.36	0.36	-83.33
lagR_plus_cueRain	4	174.38	0.86	0.23	0.59	-82.45
cue_plus_lagR	5	175.82	2.30	0.11	0.70	-81.76
lagR_x_meanT	5	176.21	2.69	0.09	0.79	-81.95
wind	3	178.02	4.50	0.04	0.83	-85.58
minT	3	178.06	4.54	0.04	0.87	-85.60
null	2	178.10	4.58	0.04	0.91	-86.84
meanT	3	178.49	4.97	0.03	0.93	-85.82
rainD	3	179.61	6.09	0.02	0.95	-86.38
lagRainD	3	179.77	6.25	0.02	0.97	-86.46
lagMeanT	3	180.01	6.49	0.01	0.98	-86.58
totR	3	180.10	6.58	0.01	0.99	-86.62
int_TR	5	181.87	8.35	0.01	1.00	-84.78

B.4.9 Adding population size to best fit model

```
obj3_pop_df <- obj3_cc |>
  left_join(
    success_df |>
      mutate(year = as.integer(as.character(year))) |>
      group_by(year) |>
      summarise(active_nests = first(active_nests), .groups = "drop"),
    by = "year"
  ) |>
  mutate(z_active_nests = as.numeric(scale(active_nests)))
```

```
summary(obj3_pop_df$active_nests)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
36.00	63.75	78.00	82.03	99.25	138.00

```
summary(obj3_pop_df$z_active_nests)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-1.9057	-0.7568	-0.1669	0.0000	0.7129	2.3171

```
m_lagR <- lm(lay_DOY_median ~ lag_total_rain_annual, data = obj3_pop_df)
m_pop <- lm(lay_DOY_median ~ z_active_nests, data = obj3_pop_df)
m_lagR_pop <- lm(lay_DOY_median ~ lag_total_rain_annual + z_active_nests,
  data = obj3_pop_df)

aictab(cand.set = list(
  lagR = m_lagR,
  pop = m_pop,
  lagR_plus_pop = m_lagR_pop
))
```

Model selection based on AICc:

	K	AICc	Delta_AICc	AICcWt	Cum.Wt	LL
lagR_plus_pop	4	170.81	0.00	0.72	0.72	-80.67
lagR	3	173.52	2.71	0.19	0.91	-83.33
pop	3	174.95	4.14	0.09	1.00	-84.05

```
dev_m2l1 <- function(mod) -2 * as.numeric(logLik(mod))
get_K <- function(mod) attr(logLik(mod), "df")
r2_fun <- function(mod) summary(mod)$r.squared

slope_se_single <- function(mod, digits = 3){
  cf <- summary(mod)$coefficients
  if (nrow(cf) == 2) paste0(round(cf[2,1], digits), " (", round(cf[2,2], digits), ")") else ""
}

build_table <- function(model_list){
  bind_rows(lapply(names(model_list), function(nm){
    mod <- model_list[[nm]]
    tibble(
      model_id = nm,
      K = get_K(mod),
```



```

    Deviance = dev_m2ll(mod),
    AICc = AICc(mod),
    `R²` = r2_fun(mod),
    `slope (SE)` = slope_se_single(mod)
  )
})) |>
  arrange(AICc) |>
  mutate(
    `ΔAICc` = AICc - min(AICc, na.rm = TRUE),
    w = exp(-0.5 * `ΔAICc`) / sum(exp(-0.5 * `ΔAICc`)),
    Deviance = round(Deviance, 2),
    AICc = round(AICc, 2),
    `ΔAICc` = round(`ΔAICc`, 2),
    w = round(w, 3),
    `R²` = round(`R²`, 3)
  )
}

model_labels <- tribble(
  ~model_id, ~Group, ~Model,

  "null", "Baseline", "Intercept only",

  "meanT", "Temperature", "Mean temperature (March)",
  "minT", "Temperature", "Minimum temperature (March)",

  "rainD", "Rain", "Rainy days (March)",
  "totR", "Rain", "Total rainfall (March)",

  "wind", "Wind", "Mean wind speed (March)",

  "lagMeanT", "Lagged effects", "Previous year mean temperature",
  "lagTotR", "Lagged effects", "Previous year total rainfall",
  "lagRainD", "Lagged effects", "Previous year rainy days",

  "cue_plus_lagR", "Joint effects", "March cues + previous year rainfall",
  "lagR_plus_cueRain", "Joint effects", "Previous year rainfall + March rainy days",

  "int_TR", "Interactions", "Mean temperature × rainy days (March)",
  "lagR_x_meanT", "Interactions", "Lag rainfall × March mean temperature"
)

group_order <- c(
  "Baseline",
  "Temperature",
  "Rain",
  "Wind",
  "Lagged effects",
  "Joint effects",
  "Interactions",
  "A posteriori models: population size",
  "Other"
)

main_df <- build_table(cand_obj3)

df <- main_df |>
  left_join(model_labels, by = "model_id") |>
  mutate(
    Group = coalesce(Group, "Other"),
    Model = coalesce(Model, model_id),
    Group = factor(Group, levels = group_order)
  ) |>
  arrange(Group, `ΔAICc`) |>
  select(Group, Model, K, Deviance, AICc, `ΔAICc`, w, `R²`, `slope (SE)`)

if (exists("m_lagR") && exists("m_pop") && exists("m_lagR_pop")) {

```

```

posthoc_models <- list(
  lagR = m_lagR,
  pop = m_pop,
  lagR_plus_pop = m_lagR_pop
)

posthoc_df <- build_table(posthoc_models) |>
  mutate(
    Group = "A posteriori models: population size",
    Model = dplyr::recode(
      model_id,
      "lagR" = "Previous year total rainfall",
      "pop" = "Number of active nests",
      "lagR_plus_pop" = "Previous year rainfall + number of active nests",
      .default = model_id
    ),
    Group = factor(Group, levels = group_order)
  ) |>
  arrange(`ΔAICc`) |>
  select(Group, Model, K, Deviance, AICc, `ΔAICc`, w, `R²`, `slope (SE)`)

df <- bind_rows(df, posthoc_df) |>
  arrange(Group, `ΔAICc`)
}

# group header rows
groups_present <- df |>
  mutate(Group_chr = as.character(Group)) |>
  pull(Group_chr) |>
  unique()

make_group_block <- function(g) {
  header <- df[0, ]
  header[1, ] <- NA
  header$Group <- g
  header$Model <- g
  header$is_group <- TRUE

  rows <- df |> filter(as.character(Group) == g)
  rows$is_group <- FALSE

  bind_rows(header, rows)
}

df2 <- bind_rows(lapply(groups_present, make_group_block))
group_rows <- which(df2$is_group)
df_print <- df2 |> select(-Group, -is_group)

ft <- flextable(df_print) |>
  font(fontname = "Times New Roman", part = "all") |>
  fontsize(size = 12, part = "all") |>
  bold(part = "header") |>
  bg(part = "header", bg = "#f2f2f2") |>
  align(j = 1, align = "left", part = "all") |>
  align(j = 2:ncol(df_print), align = "center", part = "all") |>
  bg(i = group_rows, bg = "#e6e6e6", part = "body") |>
  bold(i = group_rows, part = "body") |>
  autofit()

for (r in group_rows) {
  ft <- merge_at(ft, i = r, j = 1:ncol(df_print), part = "body")
  ft <- align(ft, i = r, j = 1, align = "left", part = "body")
}

doc <- read_docx() |>
  body_add_par(

```

```

paste0(
),
style = "Normal"
) |>
body_add_flextable(ft)

print(doc, target = "Table_Obj3_Timing_OrderedLikeObj2.docx")
cat("Saved: Table_Obj3_Timing_OrderedLikeObj2.docx\n")

```

Saved: Table_Obj3_Timing_OrderedLikeObj2.docx

```

ggplot(obj3_cc, aes(x = lag_total_rain_annual, y = lay_DOY_median)) +
  geom_point(
    size = 2.4,
    colour = "black",
    alpha = 0.9
  ) +
  geom_smooth(
    method = "lm",
    se = TRUE,
    colour = "#D5D6D2",
    fill = "#A9AAA7",
    alpha = 0.25,
    linewidth = 1.8
  ) +
  labs(
    x = "Previous-year total rainfall (mm)",
    y = "Median laying date (day of year)"
  ) +
  theme_bw(base_family = "Times New Roman") +
  theme(
    panel.grid = element_blank(),
    axis.title = element_text(face = "bold", size = 12),
    axis.text = element_text(face = "bold", size = 10)
  )

```

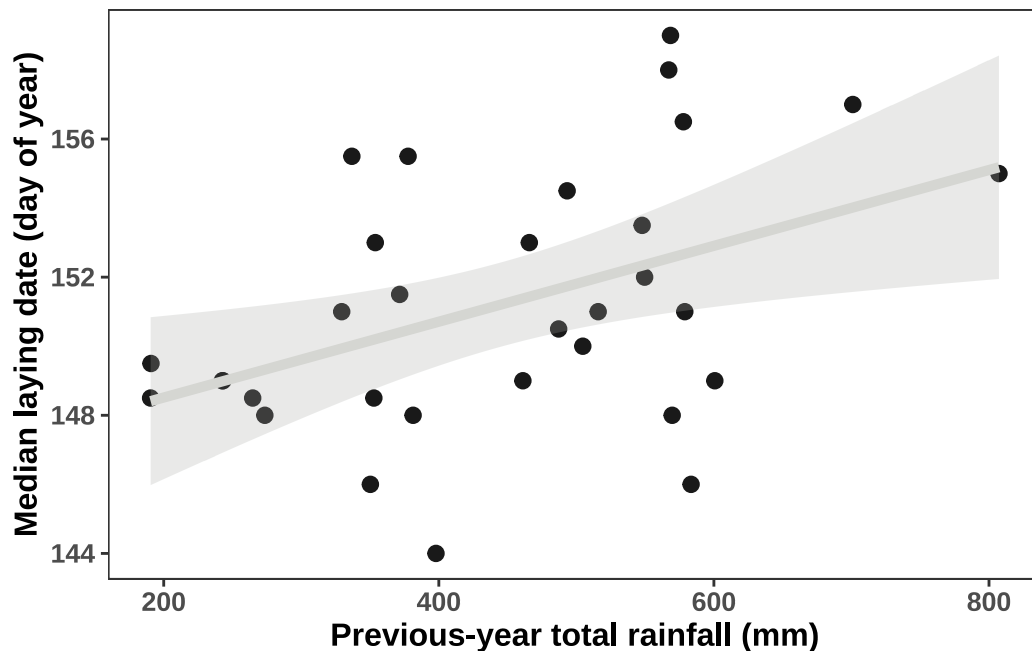


Figure 6: The relationship between the total annual rainfall in the previous year and breeding timing (median laying date)